51-DNA Sequencing Technologies

Determination of the nucleotide composition and order in a gene or genome is a foundational technology in modern medical biotechnology. DNA sequencing enables determination of the exact nucleotide sequence of a DNA fragment. DNA sequences can often reveal something about the function of the protein or RNA molecule encoded in a gene, for example, from predicted cofactor binding sites, transmembrane domains, receptor recognition sites, or DNA-binding regions. The nucleotide sequences in noncoding regions that do not encode a protein or RNA molecule may provide information about the regulation of a gene. Comparison of gene sequences among individuals can reveal mutations that contribute to genetic diseases or the relatedness among the individuals. Comparison of gene sequences among different organisms can lead to the development of hypotheses about the evolutionary relationships among organisms.

This technique was developed independently by two groups. Maxam and Gilbert invented DNA sequencing by chemical cleavage method, and Sanger developed dideoxy chain termination method for sequencing. Many different sequencing technologies have been developed, including **pyrosequencing**, sequencing using **reversible chain terminators**, and sequencing by ligation. In general, all of these methods involve (i) enzymatic addition of nucleotides to a primer based on complementarity to a template DNA fragment and (ii) detection and identification of the nucleotide(s) added. The techniques differ in the method by which the nucleotides are extended, employing either DNA polymerase to catalyze the addition of single nucleotides (sequencing by synthesis) or ligase to add a short, complementary oligonucleotide (sequencing by ligation), and in the method by which the addition is detected.

52-Dideoxynucleotide Procedure

The dideoxynucleotide procedure for DNA sequencing is based on the principle that during DNA synthesis, addition of a nucleotide triphosphate requires a free hydroxyl group on the 3′ carbon of the sugar of the last nucleotide of the growing DNA strand (Fig. A).

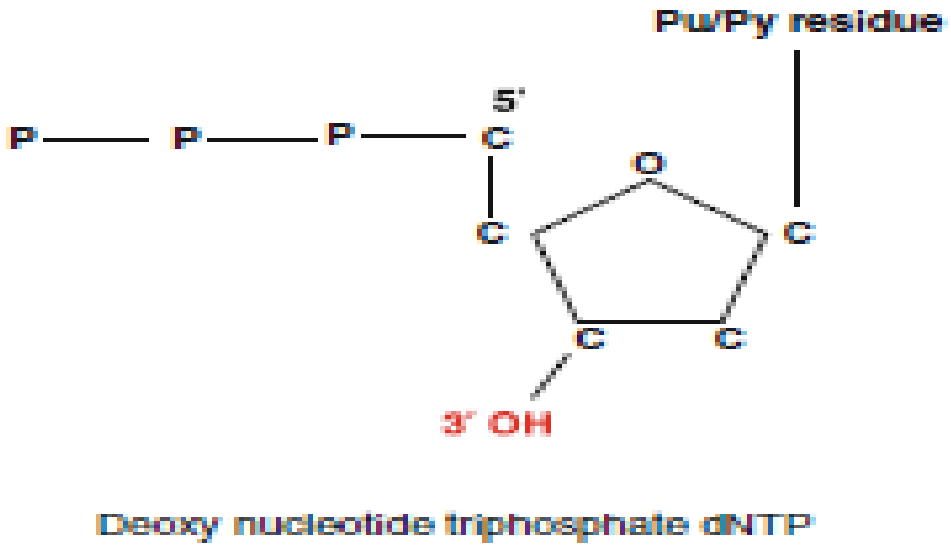


Fig-**(A)** Addition of an incoming deoxyribonucleoside triphosphate (dNTP) requires a hydroxyl group on the 3′ carbon of the last nucleotide of a growing DNA strand.

However, if a synthetic dideoxynucleotide that lacks a hydroxyl group at the 3′ carbon of the sugar moiety is incorporated at the end of the growing chain, DNA synthesis stops because a phosphodiester bond cannot be formed with the next incoming nucleotide (Fig. B).

The termination of DNA synthesis is the defining feature of the dideoxynucleotide DNA sequencing method.

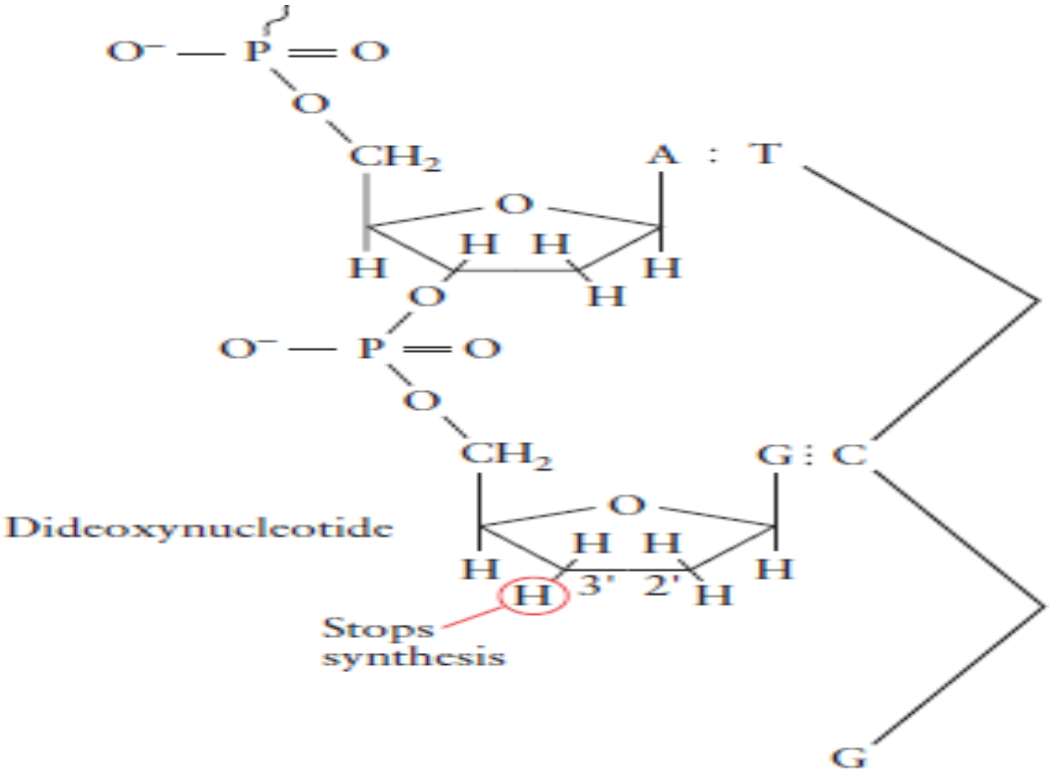


Fig- **(B)** DNA synthesis stops if a synthetic dideoxyribonucleotide that lacks a 3′ hydroxyl group is incorporated at the end of the growing chain because a phosphodiester bond cannot be formed with the next incoming nucleotide.

In a dideoxynucleotide DNA sequencing procedure, a synthetic oligonucleotide primer (~17 to 24 nucleotides) anneals to a predetermined site on the strand of the DNA to be sequenced (Fig.A). The oligonucleotide primer defines the beginning of the region to be sequenced and provides a 3′ hydroxyl group for the initiation of DNA synthesis. The reaction tube contains a mixture of the four deoxyribonucleotides (deoxyadenosine triphosphate [dATP], deoxycytidine triphosphate [dCTP], deoxyguanosine triphosphate [dGTP], and deoxythymidine triphosphate [dTTP]) and four dideoxynucleotides (dideoxyadenosine triphosphate [ddATP], ddCTP, ddGTP, and ddTTP). Each dideoxynucleotide is labeled with a different fluorescent dye. The concentration of the dideoxynucleotides is optimized to ensure that during DNA synthesis a modified DNA polymerase incorporates a dideoxynucleotide into the mixture of growing DNA

strands at every possible position. Thus, the products of the reaction are DNA molecules of all possible lengths, each of which includes the primer sequence at its 5′ end and a fluorescently labeled dideoxynucleotide at the 3′ terminus (Fig. B).

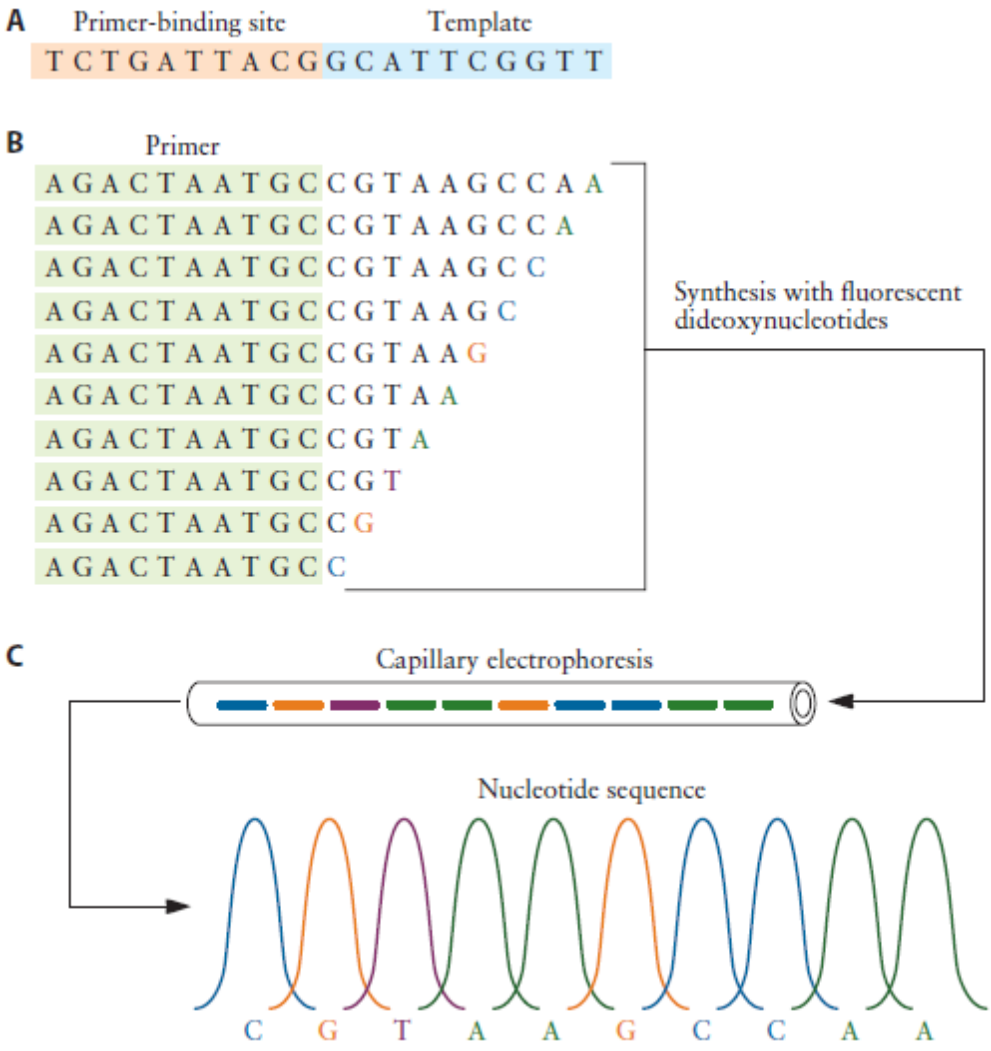


Fig- Dideoxynucleotide method for DNA sequencing. An oligonucleotide primer binds to a complementary sequence adjacent to the region to be sequenced in a single-stranded DNA template **(A)**. As DNA synthesis proceeds from the primer, dideoxynucleotides are randomly added to the growing DNA strands, thereby terminating strand extension. This results in DNA molecules of all possible lengths that have a fluorescently labeled dideoxynucleotide at the 3′ end **(B)**. DNA molecules of different sizes are separated by capillary electrophoresis, and as each molecule passes by a laser, a fluorescent signal that corresponds with one of the four dideoxynucleotides is recorded. The successive fluorescent signals are represented as a sequencing chromatogram (colored peaks) **(C)**.

PCR-based cycle sequencing is performed to minimize the amount of template DNA required for sequencing. Multiple cycles of denaturation, primer annealing, and primer extension produce large amounts of dideoxynucleotide-terminated fragments. These are applied to a polymer in a long capillary tube that enables separation of DNA fragments that differ in size by a single nucleotide. As each successive fluorescently labeled fragment moves through the polymeric matrix in an electric field and passes by a laser, the fluorescent dye is excited. Each of the four different fluorescent dyes emits a characteristic wavelength of light that represents a particular nucleotide, and the order of the fluorescent signals corresponds to the sequence of nucleotides (Fig. C). The entire dideoxynucleotide sequencing process has been automated to increase the rate of acquisition of DNA sequence data. This is essential for large-scale sequencing projects such as those involving whole prokaryotic or eukaryotic genomes. Generally, automated DNA cycle sequencing systems can read with high accuracy about 500 to 600 bases per run.

53-Pyrosequencing

Pyrosequencing was the first of the next-generation sequencing technologies to be made commercially available and has contributed to the rapid output of large amounts of sequence data by the scientific community. The basis of the technique is the detection of **pyrophosphate** that is released during DNA synthesis. When a DNA strand is extended by DNA polymerase, the α-phosphate attached to the 5′ carbon of the sugar of an incoming deoxynucleoside triphosphate forms a phosphodiester bond with the 3′ hydroxyl group of the last nucleotide of the growing strand. The terminal β- and γ-phosphates of the added nucleotide are cleaved off as a unit known as pyrophosphate (Fig. A). The release of pyrophosphate correlates with the incorporation of a specific nucleotide in the growing DNA strand. To determine the sequence of a DNA fragment

by pyrosequencing, a short DNA adaptor that serves as a binding site for a sequencing primer is first added to the end of the DNA template (Fig. 1.21B). Following annealing of the sequencing primer to the complementary adaptor sequence, one deoxynucleotide is introduced at a time in the presence of DNA polymerase.

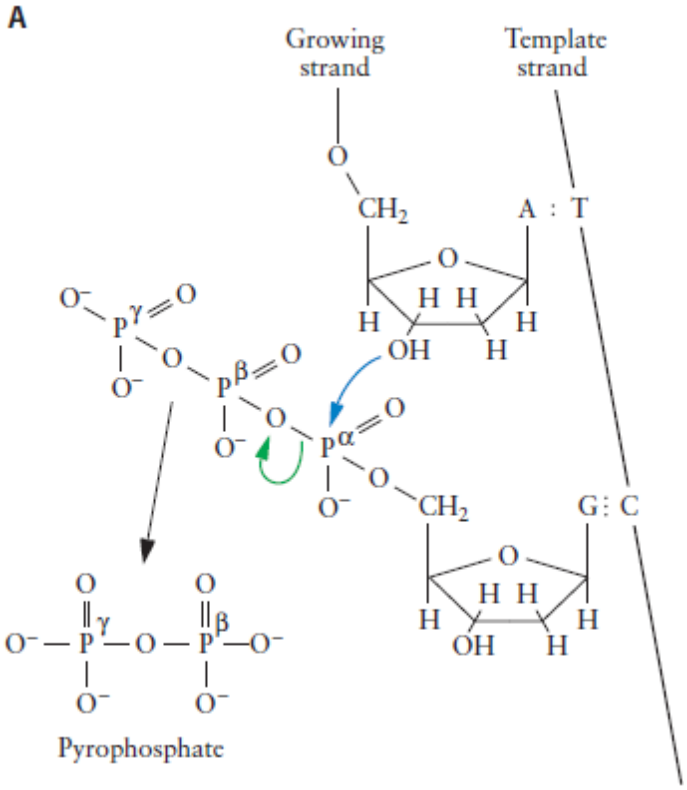


Fig- Pyrosequencing is based on the detection of pyrophosphate that is released during DNA synthesis. **(A)** A phosphodiester bond forms between the 3′ hydroxyl group of the deoxyribose sugar of the last incorporated nucleotide and the α-phosphate of the incoming nucleotide (blue arrow). The bond between the α- and β-phosphates is cleaved (green arrow), and pyrophosphate is released (black arrow).

Pyrophosphate is released only when the complementary nucleotide is incorporated at the end of the growing strand. Nucleotides that are not complementary to the template strand are not incorporated, and no pyrophosphate is formed. The pyrophosphate released following incorporation of a nucleotide is detected indirectly after enzymatic synthesis of ATP (Fig. 1.21B). Pyrophosphate combines with adenosine-5′-phosphosulfate in the presence of the

enzyme ATP sulfurylase to form ATP. In turn, ATP drives the conversion of luciferin to oxyluciferin by the enzyme luciferase, a reaction that generates light. Detection of light after each cycle of nucleotide addition and enzymatic reactions indicates the incorporation of a complementary nucleotide. The amount of light generated after the addition of a particular nucleotide is proportional to the number of nucleotides that are incorporated in the growing strand, and therefore sequences containing tracts of up to eight identical nucleotides in a row can be determined. Because the natural nucleotide dATP can participate in the luciferase reaction, dATP is replaced with deoxyadenosine α-thiotriphosphate, which can be incorporated into the growing DNA strand by DNA polymerase but is not a substrate for luciferase. Repeated cycles of nucleotide addition, pyrophosphate release, and light detection enable determination of sequences of 300 to 500 nucleotides per run (the **read length**).

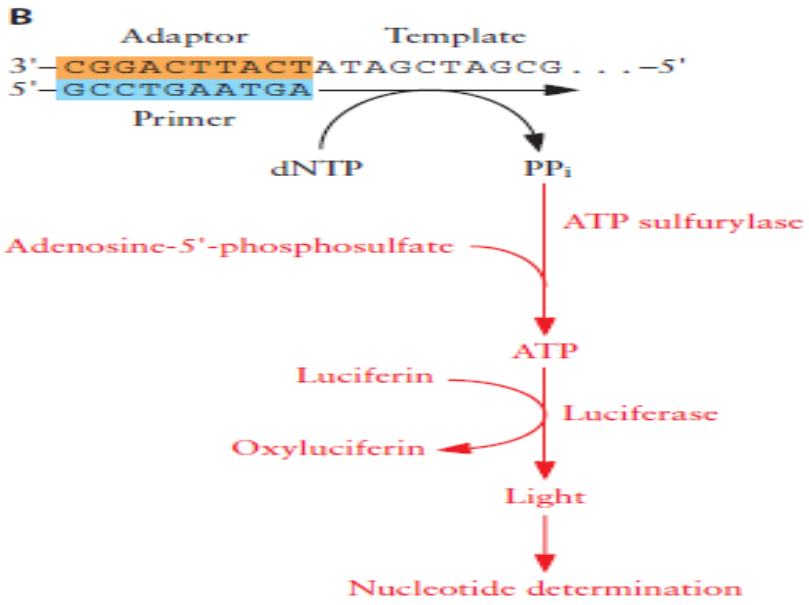


Fig- **(B)** An adaptor sequence is added to the 3′ end of the DNA sequencing template that provides a binding site for a sequencing primer. One nucleotide (deoxyribonucleoside triphosphate [dNTP]) is added at a time. If the dNTP is added by DNA polymerase to the end of the growing DNA strand, pyrophosphate (PPi) is released and detected indirectly by the synthesis of ATP. ATP is required for light generation by luciferase. The DNA sequence is determined by correlating light emission with incorporation of a particular dNTP.

54-Sequencing Using Reversible Chain Terminators

For pyrosequencing, each of the four nucleotides must be added sequentially in separate cycles. The sequence of a DNA fragment could be determined more rapidly if all the nucleotides were added together in each cycle. However, the reaction must be controlled to ensure that only a single nucleotide is incorporated during each cycle, and it must be possible to distinguish each of the four nucleotides.

Synthetic nucleotides known as reversible chain terminators have been designed to meet these criteria and form the basis of some of the next-generation sequencing-by-synthesis technologies.

Reversible chain terminators are deoxynucleoside triphosphates with two important modifications: (i) a chemical blocking group is added to the 3′ carbon of the sugar moiety to prevent addition of more than one nucleotide during each round of sequencing, and (ii) a different fluorescent dye is added to each of the four nucleotides to enable identification of the incorporated nucleotide (Fig. A).

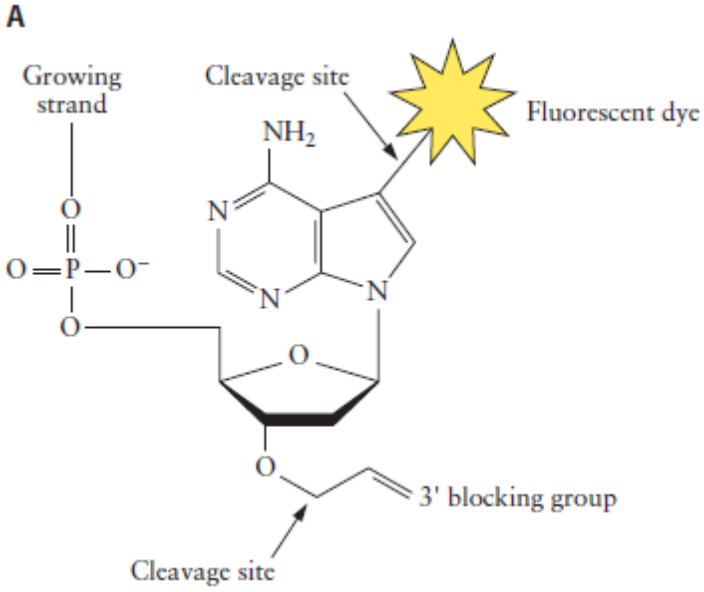


Fig-Sequencing using reversible chain terminators. **(A)** Reversible chain terminators are modified nucleotides that have a removable blocking group on the oxygen of the 3′ position of the deoxyribose sugar to prevent addition of more than one nucleotide per sequencing cycle. To enable identification, a different fluorescent dye is attached to each of the four nucleotides via a cleavable linker. Shown is the fluorescent dye attached to adenine.

The **fluorophore** is added at a position that does not interfere with either base-pairing or phosphodiester bond formation. Similar to the case with other sequencing-by-synthesis methods, DNA polymerase is employed to catalyze the addition of the modified nucleotides to an oligonucleotide primer as specified by the DNA template sequence (Fig. B).

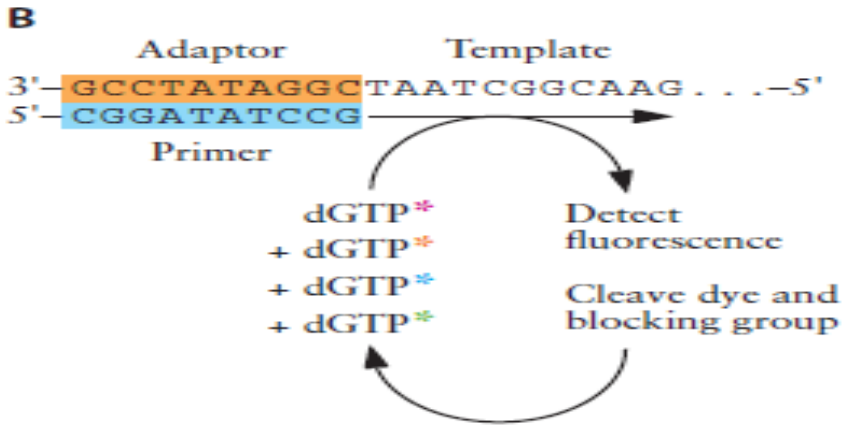


Fig-**(B)** An adaptor sequence is added to the 3′ end of the DNA sequencing template that provides a binding site for a sequencing primer. All four modified nucleotides are added in a single cycle, and a modified DNA polymerase extends the growing DNA chain by one nucleotide per cycle. Fluorescence is detected, and then the dye and the 3′ blocking group are cleaved before the next cycle. Removal of the blocking group restores the 3′ hydroxyl group for addition of the next nucleotide.

After recording fluorescent emissions, the fluorescent dye and the 3′ blocking group are removed. The blocking group is removed in a manner that restores the 3′ hydroxyl group of the sugar to enable subsequent addition of another nucleotide in the next cycle. Cycles of nucleotide addition to the growing DNA strand by DNA polymerase, acquisition of **fluorescence** data, and chemical cleavage of the blocking and dye groups are repeated to generate short read lengths, i.e., 50 to 100 nucleotides per run.

55-Sequencing by Ligation

Pyrosequencing and sequencing using reversible terminators extend the growing DNA strand by a single base during each cycle. In contrast, sequencing by ligation extends the DNA strand by ligation of short oligonucleotides, in a template-dependent fashion, and utilizes the enzyme ligase rather than DNA polymerase. In one version of this technology, the oligonucleotides are eight nucleotides in length (octamers) with two known nucleotides at the 3′ (query) end, any nucleotide in the next three (degenerate) positions, and a sequence that is common (universal) to all of the oligonucleotides at the 5′ end (Fig.A). A set of 16 different oligonucleotides representing all possible combinations of two nucleotides in the query position is used. Each oligonucleotide is tagged at the 5′ end with a different fluorescent dye that corresponds to the query nucleotide composition.



Fig-**(A)** Pools of octamers. Each octamer has two known nucleotides at the 3′ query end (colored letters), any nucleotide in the next three positions (NNN), and a universal sequence that is common to all of the octamers at the 5′ end (ZZZ). A set of 16 different octamers representing all possible combinations of two nucleotides in the query positions is shown. Each of the four octamers in a pool (each pool is shown in column) is tagged at the 5′ end with a different fluorescent dye that corresponds to the query nucleotide composition.

A short nucleotide adaptor is joined to the 5′ ends of the DNA templates that are to be sequenced (Fig. B). After denaturation of the template DNA, a primer binds to the adaptor sequence and provides a 5′ phosphate end for ligation to the 3′ hydroxyl end of an adjacent hybridized octamer. Sequential cycles of octamer ligation extend the complementary strand, and each cycle enables determination of the query dinucleotide sequence (Fig. B).

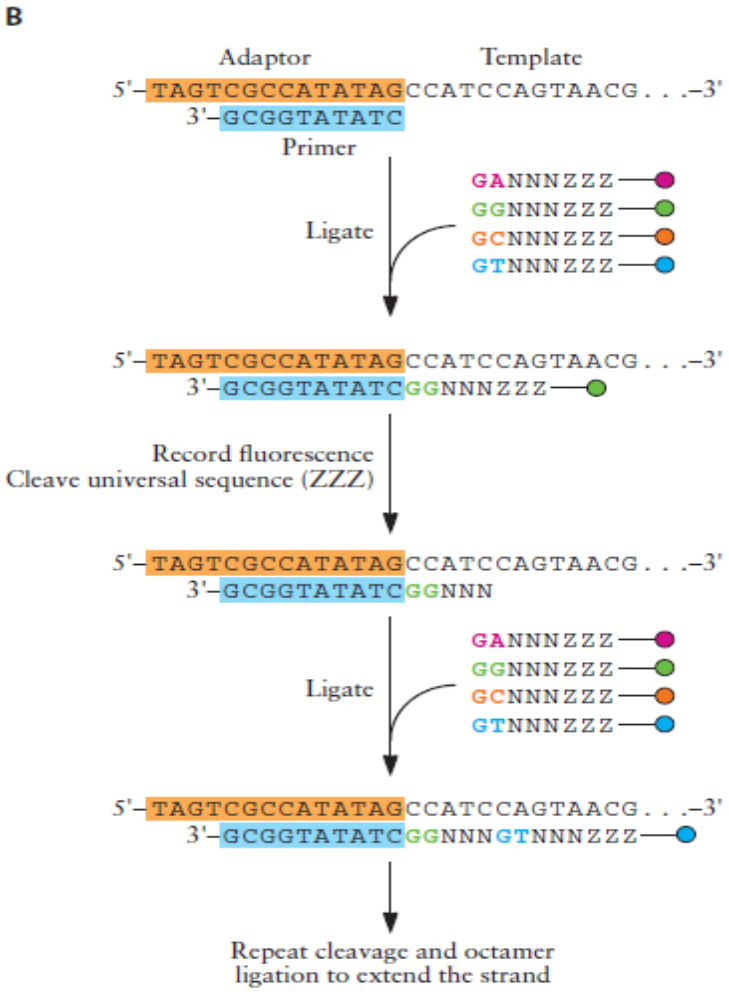


Fig-**(B)** To determine a sequence by ligation, an adaptor is first added to the 5′ end of the template DNA molecule. The adaptor provides a binding site for a sequencing primer. Pools of octamers are added, and an octamer that contains a nucleotide sequence that is complementary to the first five bases immediately adjacent to the primer will hybridize. T4 DNA ligase catalyzes the formation a phosphodiester bond between the 5′phosphate group of the primer and the 3′ hydroxyl group of the hybridized octamer. After unbound octamers are washed away, the fluorescent signal from the ligated octamer is recorded to identify the dinucleotides in the query position. The fluorescent dye is removed by cleavage of the universal sequence (ZZZ) to expose a 5′ phosphate for ligation of another octamer. The cycle of octamer ligation and dye removal is repeated to extend the DNA strand and identify discontiguous dinucleotides that are separated by three nucleotides (NNN).

One ligation cycle consists of the following steps. (i) Pools of different octamers, each tagged with a different fluorescent dye, are added along with T4 DNA ligase. Reaction conditions are set so that ligation occurs only if the bases at the 3′ end of an octamer (i.e., the query and degenerate nucleotides) are complementary to the template sequence. (ii) After washing away of

the nonligated octamers and other components, the fluorescence signal is recorded. The identity of the dinucleotides in the query position is determined by the distinctive fluorescence emitted. (iii) The fluorescent dye is removed after each cycle by cleaving the terminal universal nucleotides. Cleavage provides a free end for ligation of another octamer in the next ligation cycle.

Successive cycles of ligation extend the complementary strand and enable identification of discontiguous nucleotide pairs that are separated by three nucleotides (Fig. B). The length of the sequence read is determined by the number of ligation cycles (~50 bp). To determine the nucleotide sequence of the intervening regions (i.e., between the identified dinucleotides), the primer and octamers are removed from the template strand, and the process is repeated using a new primer that is set back on the template DNA by one nucleotide from the first primer (Fig. C). In total, the entire process of primer extension by serial octamer ligation is repeated five times, each time using a different primer that is offset from the previous primer by one nucleotide (Fig. C). In this way, the nucleotide in each position in the template DNA is identified twice in separate reactions, which increases the accuracy of the sequence determination.

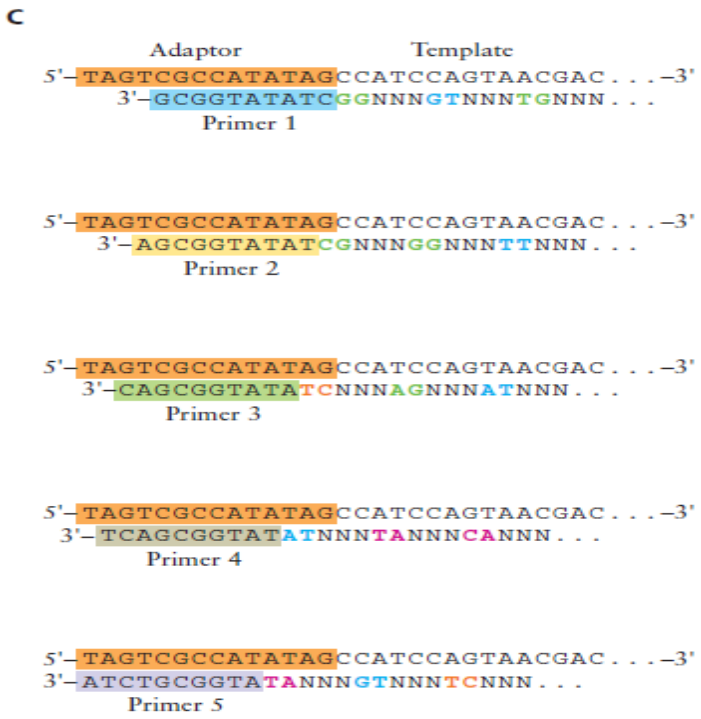


Fig-**(C)** To determine the identity of the intervening nucleotides, the primer and ligated octamers are removed and the entire process is repeated using a different primer that binds to a position on the adaptor onenucleotide from the previous primer-binding site. The process is repeated with five different primers. In this manner, each nucleotide in the template sequence is identified twice, once in two separate reactions.

56-The Immune Response

Overview of Infection and Immunity

Immunology is the study of the immune system and its responses to a host of invading pathogens and other harmful agents (e.g., insect toxins).

The **immune system** is comprised of several types of cells, tissues, and molecules that mediate resistance to infection and accompanying infectious disease, and the interaction of these cells, tissues, and molecules to infectious microbes elicits an **immune response**. Thus, the raison d'être of the immune system is to initially prevent infection and then to arrest and eliminate established infection.

The immune system is essential to maintain the health of an individual, as demonstrated by the increased susceptibility of individuals with deficient immune responses to acute, invariably life-threatening infections (Table.1). In contrast, the stimulation of immune responses against microbes by vaccination is the most effective method of protection of individuals against infection (Table 2).

Table-1 Role of the immune system in health and disease

Defensive role	Consequence
Infection	Reduced immunity increases susceptibility to infection (e.g., HIV).
	Vaccination increases immunity and protects against infections.
Immune responses to tissue transplants and novel proteins	Barriers to transplantation and gene therapy
Tumors	Possible immunotherapy for cancer

Table-2 Benefits of vaccination for some infectious diseases

Disease	Highest no. of cases (yr)	No. of cases in 2004	Decrease (%)
Diphtheria	206,939 (1921)	0	100.0
Haemophilus influenzae type b infection	20,000 (1984)	16	99.9
Hepatitis B	26,611 (1985)	6,632	75.1
Measles	894,134 (1941)	37	99.9
Mumps	152,209 (1968)	236	99.9
Pertussis	265,269 (1934)	18,957	96.8
Polio (paralytic)	21,269 (1952)	0	100.0
Rubella	57,686 (1969)	12	99.9
Tetanus	1,560 (1923)	26	98.3
TB	17,500 (1993)	14,511	96.7

Notably, by 1977, smallpox, long considered to be the most deadly and persistent human pathogenic disease, was eradicated by the World Health Organization (WHO). This was

accomplished through a massive, worldwide outbreak search and vaccination program. The underlying benefit of such immunizations is that the vaccine triggers an immune response more rapidly than the natural infection itself. The emergence of acquired immunodeficiency syndrome (AIDS) since the 1980s has further emphasized the importance of the immune system for defending individuals against infection.

Importantly, the impact of immunology extends beyond infectious disease. For example, the immune response is the major barrier to successful organ transplantation, which is being used more frequently as a therapy for organ failure. Attempts to treat cancers by stimulating immune responses against cancer cells are being tried for many human malignancies. Furthermore, abnormal immune responses are the causes of many inflammatory diseases with serious morbidity and mortality.

Antibodies, protein products synthesized by B cells during an immune response, are highly specific reagents for detecting a wide variety of molecules in the circulation and in cells and tissues. Therefore, antibodies have emerged as valuable reagents for the development of laboratory diagnostic technologies for both clinical research and clinical trials. Antibodies designed to block or eliminate potentially harmful molecules and cells are in extensive use for the treatment of immunological diseases, cancers, and other types of disorders. Accordingly, the field of immunology impacts highly on the interests of clinicians, scientists, and the lay public.

57-Functions of the Immune System

Four important functions are performed by the immune system (Table 1). First, **immunological recognition** is conducted by **leukocytes** (e.g., neutrophils, macrophages, and **natural killer cells**) of the **innate immune system** that provide an early rapid response and by the **lymphocytes** (T cells and B cells) of the later, more antigen-specific and more efficient **adaptive immune system**.

Second, to contain and eliminate the infection, **immune effector functions** that involve the complement system of blood proteins, enzymes, and antibodies together with certain T cells and B cells are required.

Third, while combating foreign pathogens, the immune system must learn not to elicit damage to self-tissues and other components of the body. It accomplishes this by **immune regulation**, a process in which regulatory lymphocytes control various immune responses against self-components. A failure in immune regulation could potentially result in allergy and autoimmune disease.

Fourth, the immune system must be equipped to protect an individual against recurring disease upon exposure to the same or a closely related pathogen. This task is uniquely mediated by the adaptive immune system during a response known as **immunological memory**. After exposure to an infectious pathogen, a person will make a rapid and more vigorous response to the pathogen upon subsequent encounter with this pathogen and will acquire protective immunity against it. Thus, a very significant challenge facing immunologists today is the identification of relevant cellular pathways and molecule that generate long-lasting immunity (i.e., memory) to pathogens, with the aim of identifying novel **immunotherapies** using cells and/or antibodies that eradicate such pathogens.

Table-1 Functions of the immune system

Function	Cell type
Immune recognition	Leukocytes (neutrophils, macrophages, and natural killer cells) and lymphocytes (T cells and B cells)
Contain and eliminate infection	Effector cells (T cells and B cells) and molecules (complement proteins, enzymes, and antibodies)
Immune regulation	Regulatory lymphocytes
Immunological memory	Memory lymphocytes

Two major types of immunity are carried out by cells of the immune system in response to inflammation and infection (Fig.1). First, **innate immunity** provides a mechanism of host defense that mediates the initial protection against microbial infections. Second, **adaptive immunity**, which develops more slowly and confers specificity against a foreign antigen, mediates the later and more vigorous defense against infections.

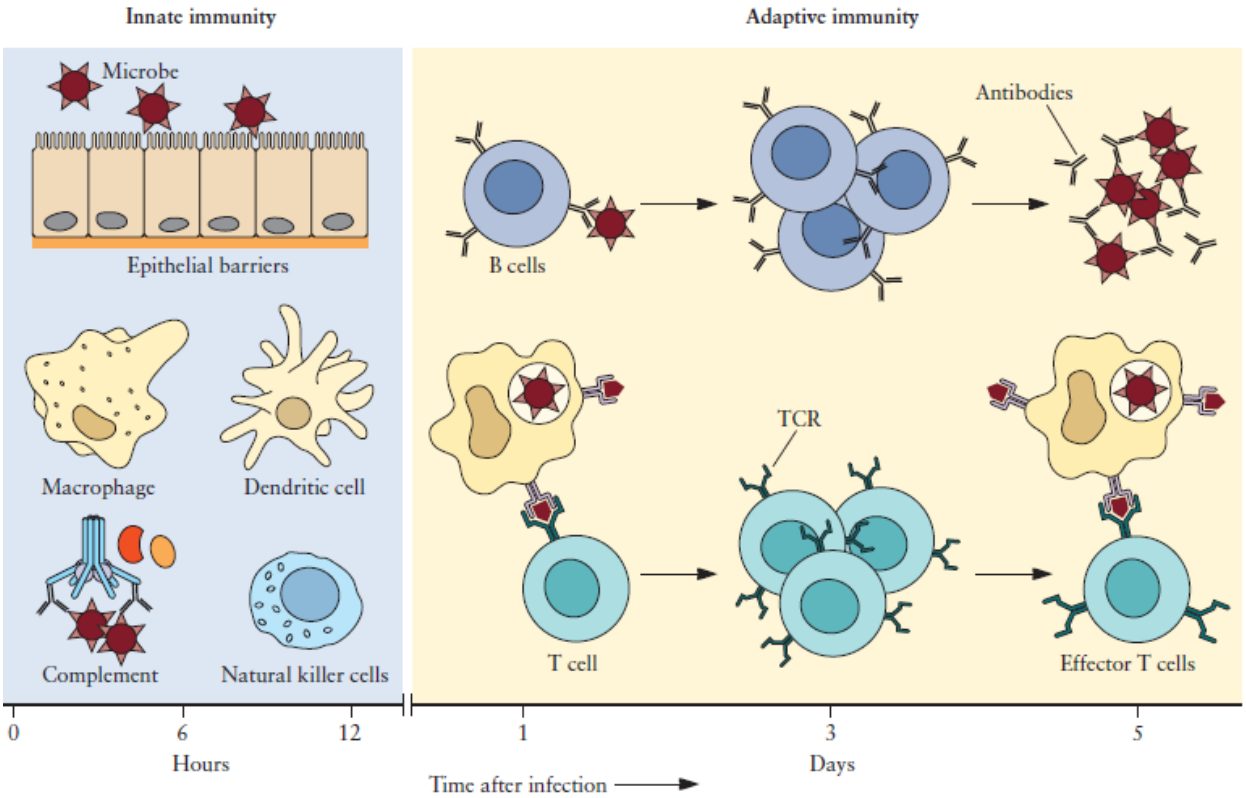


Fig-1 Mechanisms of innate and adaptive immunity. The initial defense against infections is mediated by mechanisms of innate immunity that either protect from infection (mucosal epithelial barriers) or eliminate microbes (macrophages, natural killer [NK] cells, and complement system). Adaptive immune responses develop later and are mediated by lymphocytes and their products. While antibodies block infections and eliminate microbes, T cells eradicate intracellular microbes. The kinetics of innate (early after infection) and adaptive (late after infection) immune responses are different and may vary with the type of infection incurred.

59-Cellular network of innate immune system

The white blood cells (WBCs), identified as leukocytes,are different from other cells of the body because they are normally associated with a particular organ. It has been reported that leukocytes are able to move freely in the body and network with cellular debris, foreign particles, and invading pathogenic microorganisms. Interestingly, most innate leukocytes cannot multiply or reproduce on their own, unlike other body cells. These leukocytes are the products of hematopoietic stem cells present in the bone marrow and we briefly describe the various types of innate leukocytes.

Mast cells

In the human body, mast cells are basically innate immune cells that reside in the connective tissue and in the mucous membranes, and these cells are closely associated with defense against pathogens and wound healing. After activation, mast cells speedily release some granules that are equipped with histamine and heparin and chemotactic cytokines into the host cell environment. It has been reported that histamines open blood vessels, causing irritation or inflammation, and recruit neutrophils and macrophages to defend the host cells (Figure 1).

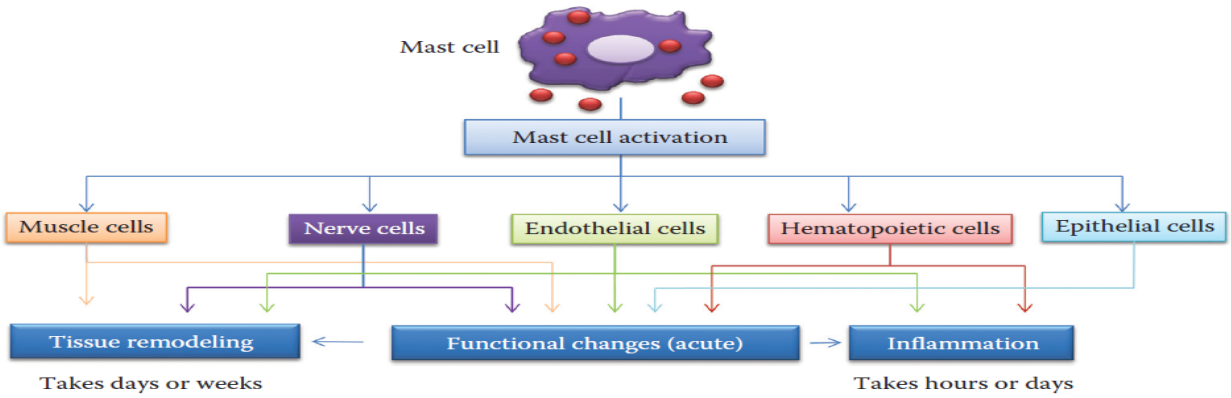


Fig-1 Mast cells and their role in innate and acquired immunity.

Phagocytes

Phagocytes are basically immune cells that engulf pathogens or particles to protect the body from any infection. It has been reported that phagocytes engulf a particle or pathogen by extending parts of its plasma membrane and wrapping the membrane around the particle until it is completely enveloped and covered. Once the pathogen is completely covered, the lysosome, which contains enzymes and acids, starts killing and digesting the foreign particle or organism. It has been described that phagocytes generally patrol the human body searching for pathogens, and are also able to respond to molecular signals produced by other cells, which are called cytokines. There are various types of phagocytic cells reported in the human immune system that include neutrophils, macrophages, and dendritic cells (Figure 2).

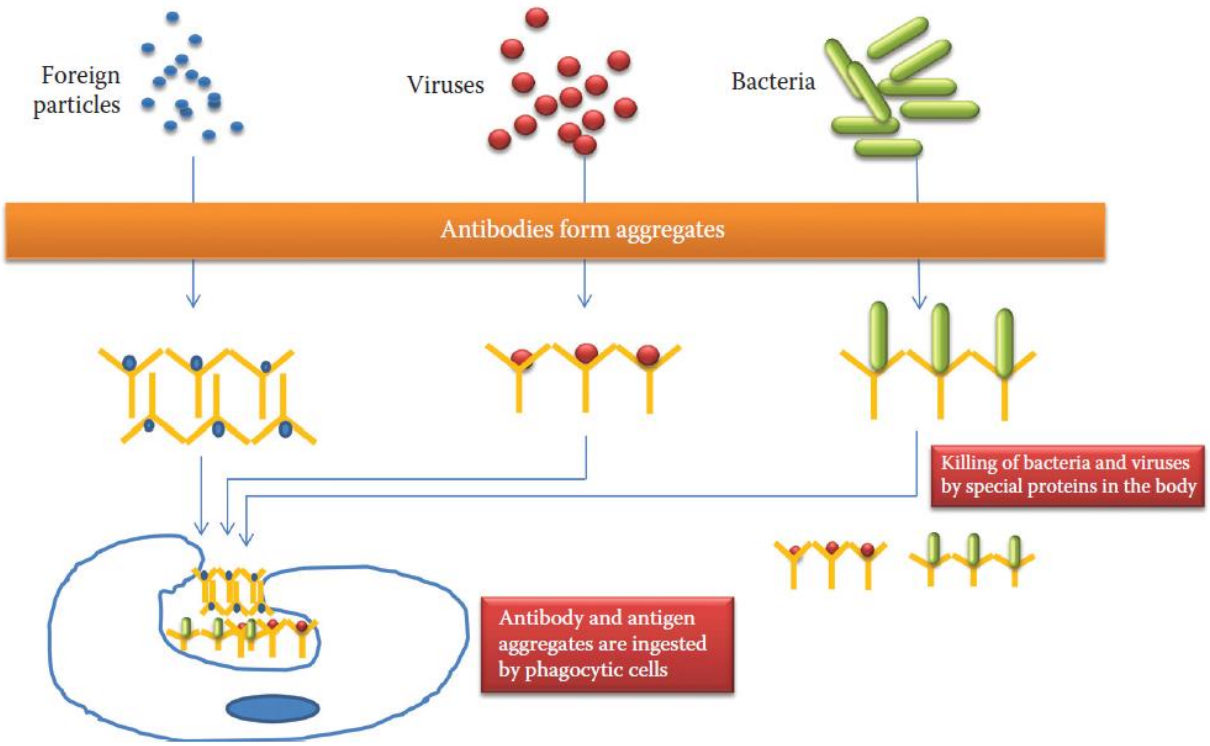


Fig-2 Role of phagocytic cells in immune protection.

60-Macrophages

Macrophages are large leukocytes that are able to move outside the vascular system and enter the areas between cells in the hunt of invading pathogens. It has been reported that these macrophages are the most efficient phagocytes, and can kill large numbers of bacteria or other cells or microorganisms. The attachment of anti-bacterial molecules to receptors of a macrophage terminates the bacterial population through respiratory spurt, causing the release of reactive oxygen species. It has been reported that pathogens can also stimulate the macrophage to produce chemokines, which command other cells to the site of infection.

Neutrophils

Because of the presence of granules in their cytoplasm, neutrophils, eosinophils, and basophils are known as granulocytes. Neutrophils are also known as polymorphonuclear cells (PMNs) due to their typical lobed nuclei. It has been reported that neutrophil granules contain a variety of toxic elements that destroy or inhibit the growth of bacteria and fungi. Like macrophages, neutrophils attack pathogens by activating a respiratory spurt that contains oxidizing agents, including free oxygen radicals, hydrogen peroxide, and hypochlorite. It has been reported that neutrophils are the most plentiful type of phagocyte, normally representing 50–60% of the total mobile leukocytes, and they are usually the first type of cells to reach the site of an infection. It has been estimated that the bone marrow of a healthy adult human being produces more than 100 billion neutrophils per day, and produces more than 10 times more neutrophils during acute inflammation or infection.

Dendritic cells

Dendritic cells are phagocytic cells present in human tissues that are in contact with the external atmosphere, primarily the skin, where they are referred to as Langerhans cells, and the inner mucosal lining of the nose, stomach, intestines, and lungs. The structure of dendrite cells closely resembles neuronal dendrites; that is why they are called dendritic cells. They have no association with neuronal cells. It has been suggested that dendritic cells are very vital in antigen performance and assist as a link between the innate and adaptive immune defense system (Figure 3).

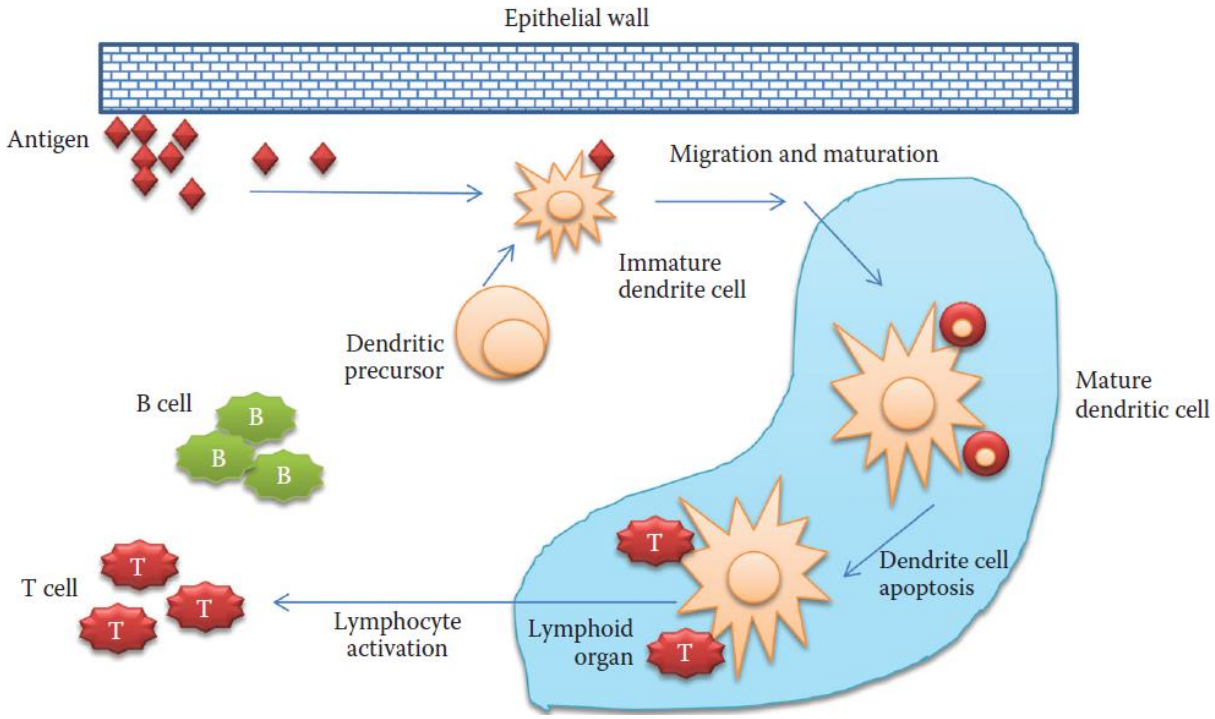


Fig-3 Dendritic cells and their biological actions.

Basophils and eosinophils

Basophils and eosinophils are known to relate to the neutrophils. Moreover, the basophils release histamine upon parasite entry, which are critical in the defense against parasites, and are also involved in allergic reactions. These basophils during activation secrete a highly toxic protein and free radicals that instantly kill bacteria and parasites; however, they are also known to cause cell and tissue damage during allergic-related reactions.

Natural killer cells

Natural killer (NK) cells are an important member of the innate immune defense system that does not attack invading pathogens or microbes. In contrast to phagocytes, NK cells can terminate compromised host cells, such as cancerous cells or virus-infected cells. They are called NK cells because of an initial idea that these NK cells do not require activation to kill cells (Figures 4).

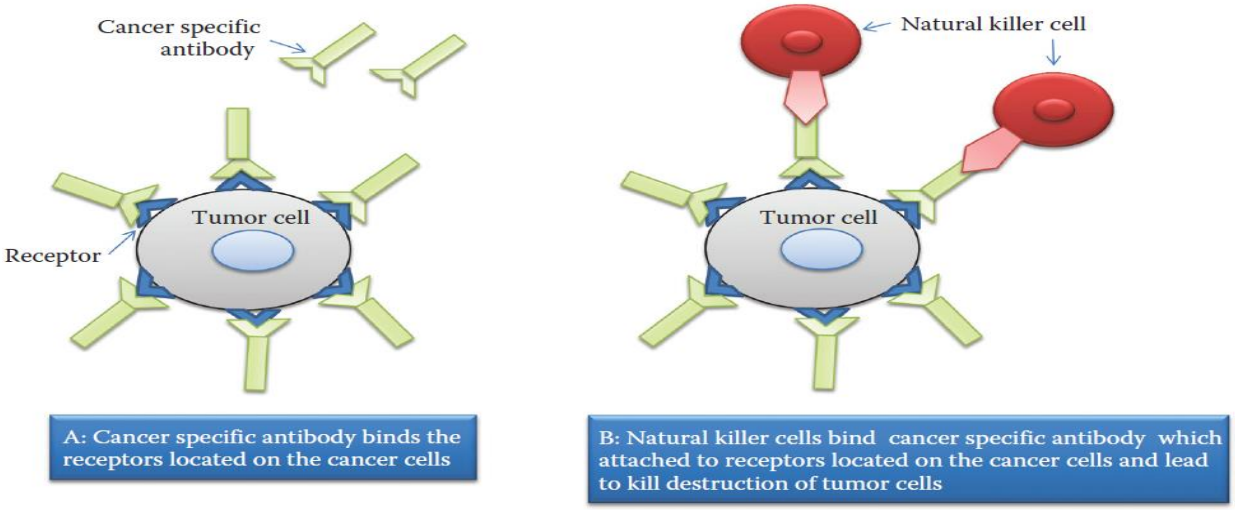


Fig 4- NK cell destroying cancerous cells.

61-Adaptive immune system

The adaptive immune system is basically a defense system in the human body that remembers the entry and action of each and every pathogen. Moreover, adaptive immune response is antigen specific and needs the recognition of specific antigens during antigen presentation. Antigen specificity can allow the generation of responses that are tailored to specific pathogens or pathogen-infected cells in the body. The special adaptive capability of immune cells is maintained in the body by memory cells.

Lymphocytes

The lymphocytes belong to the adaptive immune system. There are two different types of lymphocytes, B cells and T cells. These cells are generally derived from hematopoietic stem cells in the bone marrow. Moreover, B cells take part in the humoral immune response, while T cells are mainly involved in cell-mediated immune response. Both B and T cells contain receptor molecules that distinguish specific targets. For example, T cells recognize a pathogen only after antigens are processed and presented in combination with major histocompatibility complex (MHC) molecule. There are two major subtypes of T cells—killer T cells and helper T cells. Killer T cells can only identify antigens coupled to Class-I MHC molecules, while helper T cells only identify antigens coupled to Class-II MHC molecules. A third, minor subtype is the gamma–delta T cells that distinguish integral host antigens that are not attached to MHC receptors.

62-Killer T cells

Killer T cells are a subtype of T cells that can either kill cells that are infected with viruses and pathogens, or make them dysfunctional. It has been shown that killer T cells are activated when

T cell receptor binds to antigens of another MHC cell. Moreover, the recognition of this MHC–antigen complex is aided by a coreceptor localized on the T cell, which is also called CD-8. It has been suggested that T cells travel throughout the body in hunt of cells where the MHC-I receptors contain this antigen. As soon as an activated T cell makes contact with such cells, it releases cytotoxins that form holes in the target cell's plasma membrane, and subsequently allow ions, water, and toxins to enter inside the cell. Furthermore, the entry of another toxin called *granulysin* (a protease) persuades the target cell to undergo apoptosis. More significantly, destroying host cells by T cells is particularly important in preventing the replication of viruses.

Helper T cells

The activation of helper T cells can result in the release of cytokines as well as stimulation of macrophages such as killer T cells and B cells. The stimulation of B cells and macrophages induces a proliferation of T helper cells. These T cells regulate both the innate and adaptive immune responses and help the body to decide which kinds of immune responses to make toward a specific pathogen. It has been reported that these helper T cells have no cytotoxic activity and do not kill infected cells (Figure 5.9). Helper T cells express T cell receptors that identify antigens attached to Class-II MHC molecules. It has been reported that helper T cells have a weaker association with the MHC–antigen complex than killer T cells, which means many receptors located on the helper T cell can be attached by an MHC antigen to activate the helper cell. Moreover, helper T cell activation also requires longer duration of assurance with an antigen-presenting cell. In addition, helper T cell activation results in upregulation of molecules expressed on the T cell surface, which provide extra stimulatory signals normally required to activate antibody-producing B cells.

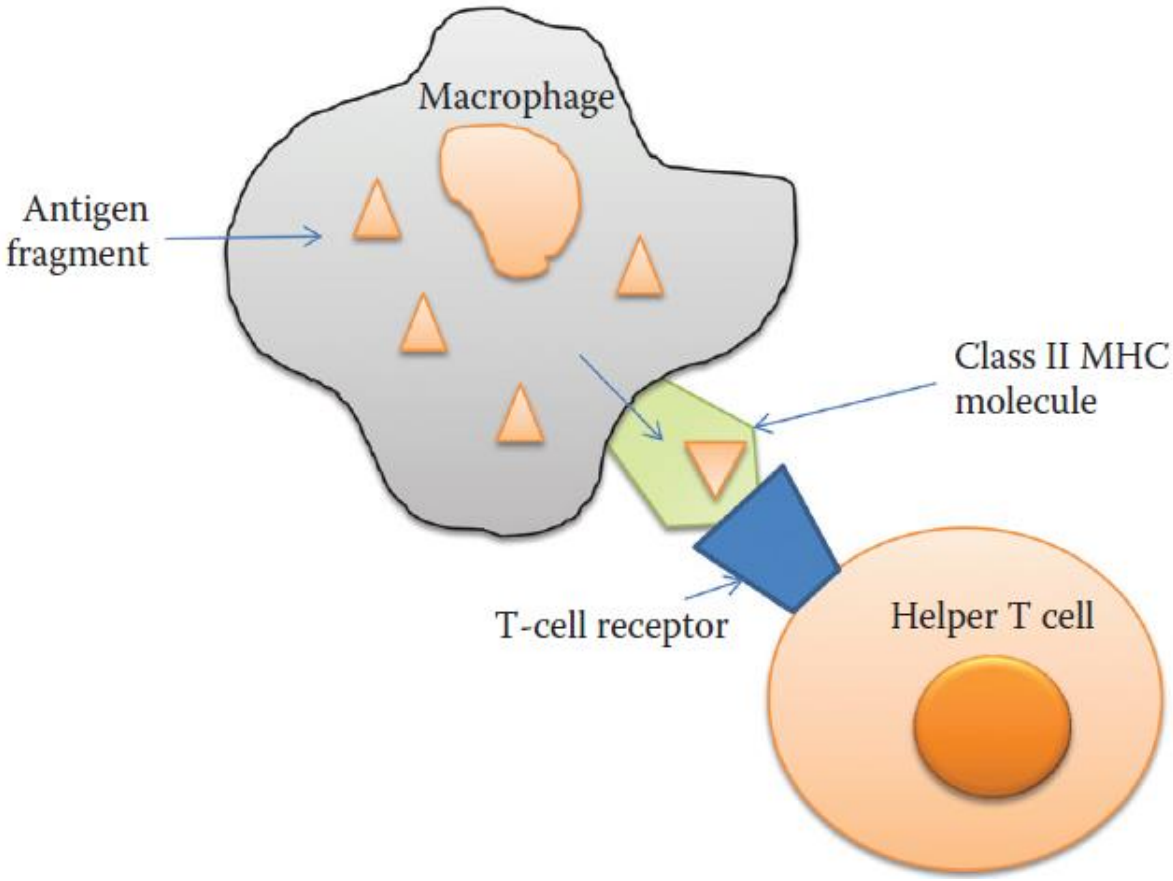


Fig-1 Function of T helper cells.

63-B lymphocytes and antibodies

The B lymphocytes and antibody cells detect pathogens when the antibodies on their surface attach to a specific foreign antigen. It has been described that this antigen–antibody complex is taken up by the B cell and processed by proteolysis into peptides. Later on, the B cells display these antigenic peptides on their surface of MHC Class-II molecules. As soon as B cells are activated, they begin to multiply, and their offspring discharges millions of copies of the antibody that recognizes this antigen. Furthermore, these antibodies can circulate in blood and stick to pathogens expressing the antigen, destroying these pathogens with phagocytes.

Immunological memory

When B cells and T cells are activated and begin to reproduce, some of their offspring can become long-lived memory cells. This process is called immunological memory. It has been suggested that in all organisms, these memory cells can remember each and every specific pathogen encountered and can enhance a strong response if the pathogen is detected or enters again. This kind of immunological response is basically adaptive in nature, because it occurs during the lifetime of an individual. It has been reported that immunological memory can be classified as passive short-term memory or active long-term memory.

64-Passive memory

It has been reported that newborn infants have no prior exposure to pathogens and are particularly vulnerable to infections. During this critical period of child development, the mother provides immunity against any kind of infection by producing a particular type of antibody known as immunoglobulin-G, which is transported into the baby directly across the placenta. These antibodies help the infant to fight against any infection. Furthermore, mother's breast milk contains antibodies that are transferred to the infant and protect against bacterial infections until the newborn body can produce its own antibodies. This type of development of immunity between the mother and child is called passive immunity because the fetus does not make any antibodies on its own, it only borrows them from the mother. It has been reported that passive immunity is only for a short duration of time, up to several months. Interestingly, now, it is possible to induce protective passive immunity in the infant by artificial ways.

Active memory and immunization

It has been suggested that long-term active memory is acquired following infection by activating B cells and T cells. Active immunity can be produced by vaccination. The main principle behind vaccination is called immunization where a patient is injected with an antigen developed from a nonactive or killed form of pathogens to inspire the body to develop specific immunity against that particular pathogen.

65-Immune diseases

The human immune system is remarkable in protecting the body from illness; however, under disease conditions, the immune system may fail to defend the body against various infections. This is called an immune-compromised immune system. This condition makes the body vulnerable to various viral, bacterial, or fungal infections. The immune-related diseases can be classified into three broad categories such as immunodeficiency, autoimmunity, and hypersensitivities.

Immunodeficiency

Immunodeficiency is a physiological condition where the immune system's capacity to fight infectious disease has become dysfunctional or entirely absent. In most cases, immunodeficiencies are acquired diseases, but some people are born with defects in the immune system, which is known as a primary immunodeficiency. It has been reported that a person who has an immunodeficiency of any kind is considered to be immune-compromised and such a person may be particularly vulnerable to opportunistic infections, in addition to normal infections that could affect everyone.

Primary immunodeficiency

A number of rare diseases have been reported susceptible to infections from childhood onward and many of these diseases are hereditary and autosomal recessive or X-linked in nature. There are more than 80 primary immunodeficiency syndromes reported in humans and they are

generally grouped by their malfunctioning of lymphocytes or granulocyte cells. The treatment of primary immunodeficiencies depends on the nature of the defect, and may involve antibody infusions, long-term antibiotics, and stem cell transplantation.

Acquired immunodeficiency

An immune deficiency that is caused due to external processes is called secondary or acquired immunodeficiency. The common causes for secondary immunodeficiency are malnutrition, aging, and particular medications such as chemotherapy, disease-modifying antirheumatic drugs, and glucocorticoids, which are immunosuppressive drugs after organ transplants. It has been reported that there are many specific diseases that directly or indirectly impair the immune system and these include many types of cancer, predominantly those of the bone marrow and blood cells, such as lymphoma, leukemia, multiple myeloma, and certain chronic infections. Immunodeficiency is also the hallmark of AIDS, caused by HIV.

66-Autoimmunity

In a healthy person, immunity generally protects the person from all infectious diseases, but in some individuals, it works against itself and destroys the person's immune system. This condition is generally known as an autoimmune disease. It has been suggested that any disease fallouts from such an abnormal immune response are called an autoimmune disease. Autoimmunity is frequently caused by a deficiency of germ cell development that results in the development of a condition where body cells work against their own cells and tissues. There are several examples of autoimmunity disease such as celiac disease, Churg–Strauss syndrome,

Hashimoto's thyroiditis, Graves' disease, idiopathic thrombocytopenic purpura, diabetes mellitus type 1, systemic lupus erythematosus, Sjögren's syndrome, and rheumatoid arthritis (RA).

Classifications of autoimmune diseases

Autoimmune diseases can be classified into systemic autoimmune disease and localized autoimmune disorders, depending on the clinical and pathological features of each disease. The examples of systemic autoimmune diseases are Sjögren's syndrome, scleroderma, RA, systemic lupus erythematosus, and dermatomyositis. It has been proposed that these disease conditions incline to be associated with autoantibodies to antigens, which are not tissue specific. The names of affected organs/tissues with autoimmune disease are listed in Table .1.

Table-1 Name of Syndromes That Affect Various Body Organs

Name of syndrome	Affected organ/tissue
Diabetes mellitus type 1	Pancreas
Celiac disease, pernicious anemia	Intestine
Pemphigus vulgaris, vitiligo	Skin
Autoimmune hemolytic anemia, idiopathic thrombocytopenic purpura	Blood
Myasthenia gravis	Brain

67-Classifications of cell transplantation

In cell transplantation, the cells or tissues can be classified based on their utilization in different parts of the body. We have given a few examples where cell-based transplantation is effectively used in clinical research.

Autologous cells

The autologous cell transplantation is basically a transplantation in which cells obtained from patients can be transplanted into the same patient. One significance of the autologous transplantation is that there is a high rate of success as the transplanted cells easily can be accepted by the patient's body and there are few problems associated with ejection and pathogen transmission. There are however, a few shortcomings in the autologous cell transplantation as the availability of required cells to be transplanted are found in very low in numbers and also genetic disease-suitable autologous cells are not available. Another disadvantage of autologous cell transplantation is that older patients do not have sufficient numbers of autologous cells to be used for cell transplantation. The old patients need to undergo a surgical operation in order to isolate the cells which might lead to donor site infection or chronic pain. Autologous cells also must be cultured from samples before they can be used and these are lengthy procedures, so autologous cell transplantation might take a long time. In recent times there has been a trend toward the use of human mesenchymal stem cells (MSCs) isolated from bone marrow and fat cells. These MSCs can be differentiated into a variety of tissue types including bone, cartilage, fat, and nerve cells. A large number of cells can be easily and quickly isolated from a fat source, consequently opening the potential for large numbers of cells to be quickly and easily obtained.

68-Allogeneic cells

Allogeneic cells come from the body of a donor of the same species; for example, the cells from one person (donor) can be used by another person (patient). There are some ethical restraints on the use of human cells for *in vitro* studies, however, the use of dermal fibroblasts from human foreskin has been demonstrated to be clinically safe and consequently a viable choice for tissue engineering of skin.

Xenogeneic cells

The cells which are isolated from two different species are called as xenogeneic cells; for example, the cells from one species (sheep) can be used in another species (human). In particular, animal cells have been widely used in experiments aimed at the construction of cardiovascular implants which later can be used in humans.

Syngeneic or isogenic cells

This type of cell isolates from genetically identical individuals, such as twins, clones, or extremely inbred research animal models.

69-ORGAN TRANSPLANTATION

ORGAN TRANSPLANT SUCCESSES AND FAILURES

It has been over 50 years since the first successful human organ transplant was done—a kidney transplanted from one identical twin to another—and the procedure has become almost routine. The list of organs transplanted includes not just kidneys, but hearts, livers, lungs, and the pancreas.

The number of transplants performed is impressive, considering both how complicated the surgery is and how strong a fight the immune system launches against the new organ. More than 325,000 transplants were performed in the United States from 1988 through August 2004. The

average number per year is now about 25,000. Kidney transplants account for more than half of all transplants; liver transplants make up about 20%, hearts 10%, and lungs 5%. The number of transplants may seem large, but an even bigger number includes those people who are on a waiting list for an organ—more than 86,000 people right now.

Because human beings have two kidneys and can function just fine with only one, kidney transplants can come from living donors. In recent years, methods to transplant just a part of a liver have succeeded because the liver can repair and regenerate itself. A few hundred people each year, often children, receive a lobe of a liver from a family member.

Human-to-human organ transplantation has risks. The human immune system is finely tuned to recognize and destroy invaders, and because it carries different forms of transplantation markers from those of the recipient, the transplanted organ is seen as an invader, unless the donor and recipient share exactly the same genes, as in that first successful kidney transplant between identical twins. A major risk exists if antibodies to the donor cells are present in the recipient's blood at the time of transplant. If donor-reactive antibodies are present when the blood connection to the organ is made during surgery, the reaction against the new organ may be so rapid that the blood vessels are blocked and the organ cells die. This rapid antibody reaction to the cells lining the organ blood vessels is called **hyperacute rejection**. Tests for antibodies lurking in the recipient's blood that could cause hyperacute rejection are performed before the transplant to determine who receives a donated organ. The ABO blood group system, critical in blood transfusion, is also important for some, but not all, transplanted organs. An organ donor must be a suitable ABO type for the patient or ABO antibodies in the recipient's blood will cause hyperacute rejection.

After the surgery, new antibodies and killer lymphocytes that cause rejection may develop within days. To prevent this from happening, or at least reduce the chances of it, scientists have discovered and developed a number of **immunosuppressive drugs** that help extend the life of the transplanted organ and, thus, the life of its recipient. **Corticosteroids** and the cancer chemotherapy drug azathioprine were the first drugs used to suppress the immune system for organ transplants. Although their long-term use can have serious side effects, steroids such as prednisone are still widely used. In the 1970s, the introduction of **cyclosporine**, the product of a soil fungus, dramatically improved one-year graft and transplant patient survival. Other widely used chemical immunosuppressive drugs include tacrolimus, mycophenolate mofetil, and sirolimus. Several monoclonal antibodies that remove or shut down T lymphocytes (such as muromonab, daclizumab, and basiliximab) are used to prevent or treat rejection. A preparation of animal antibodies raised against human lymphocytes is still used for the same purpose. Despite all these tools, slow or chronic rejection of the organ remains an ongoing problem. The success of organ transplantation is measured by organ and patient survival. Over 80% of kidney patients and 65% of their grafts survive for five years.

TAKING ORGANS FROM OTHER ANIMALS

Because the transplant waiting list continues to grow, scientists have explored nonhuman animals as a possible source of organs. Despite the shock and repulsion it sometimes causes, the idea of **xenotransplantation**, or transplantation of an organ from one species to another, deserves some serious consideration.

70-Transgenic organisms

Transgenic organisms are genetically modified organisms (GMOs) which have inserted DNA of different species origin. The inserted DNA/gene is known as transgene. Transgene is introduced into the organism, using recombinant DNA technology. The gene must be transmitted through the germline so that every cell, including germ cells, of the body contains the same modified genetic material.

Basic Requirements

In gene transfer, the knowledge of the biochemical and physiological mechanisms of its action, regulation of gene expression, and safety of transgene and its product is important. The process requires some basic steps for successful transfer:

- The extraction of nucleic acid (DNA/RNA). It is the first step in the genetic engineering process.

- After DNA extraction and selection of suitable gene, the next step is gene cloning . Gene cloning has four stages: (1) generation of DNA fragments, (2) ligation to a vector, (3) propagation in a host cell, and (4) selection of the transformed host.

- Gene is designed in such a way that it is under appropriate promoter and correct regulatory elements. Selectable marker gene may be incorporated for selection of the transformants from non-transformants.

- Incorporation of reporter gene allows detection of gene expression in transformed cells and enhancers are essential for efficient transcriptional control.

- The gene of interest (GOI) is ligated to selectable marker gene and reporter gene. It is helpful for screening, detection, and identification of the transgene containing host cell. Reported genes may be β-glucuronidase gene (GUS) or green fluorescent protein or luciferase gene. GUS produces blue product when it acts on its substrate, and thus cells appear blue, whereas GFP and luciferase transformed cells emit light when excited with appropriate wavelength of light.

71-Production of Transgenic Animals

There are two basic strategies for producing transgenic animals. These are the so-called “gain of function” or “loss of function” transgenics. The “gain of function” relies on addition of a cloned fragment of DNA to an animal’s genome to obtain new expression of a gene product that did not previously exist in that cell or tissue type.

“Loss of function” is a more direct way to determine the function of a lost/inactive gene after its alteration (Fig. 1) and then observe the physical manifestation (phenotype) of that genetic trait, for example, the mouse is obese or diabetic. These types of mice are often called knockouts, and from several years now, it has been possible to generate “ knockout” mice , which carry specifically defined mutations in the gene of interest using gene targeting in mouse embryonic stem cells .

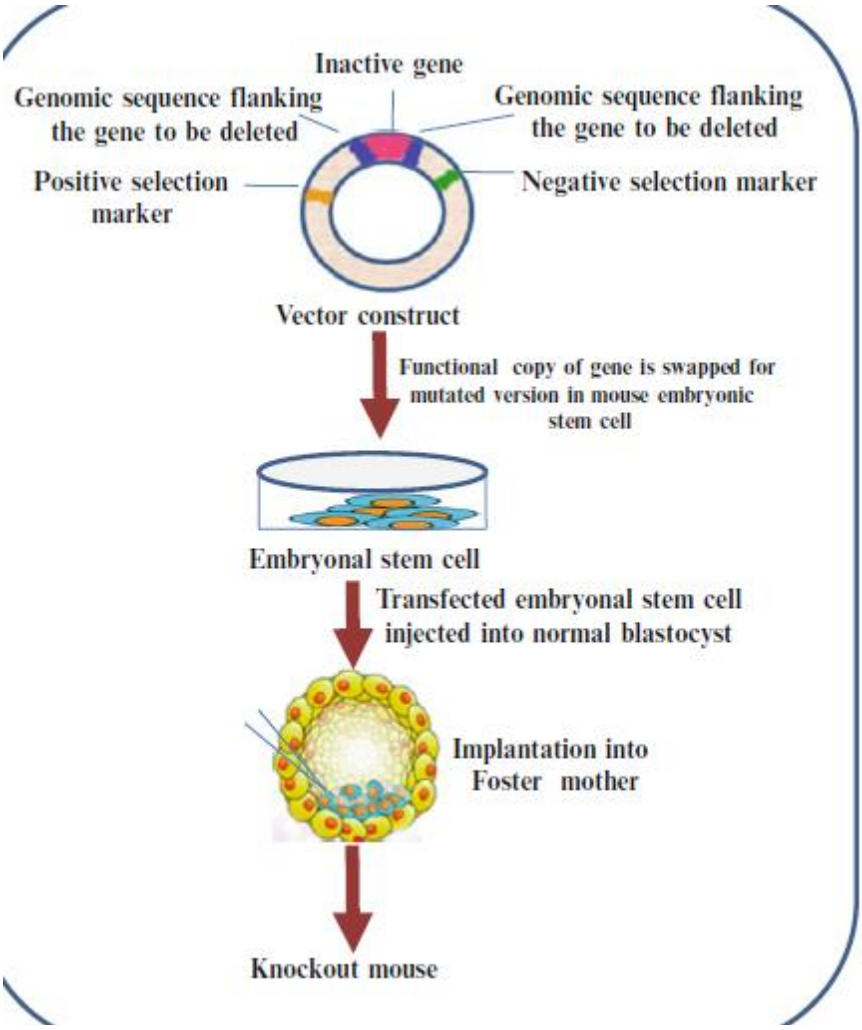


Fig-1 The figure shows the loss of function cassette where inactive gene is incorporated for homologous recombination with genomic sequence flanking the target gene with positive and negative selection marker gene. The vector exchanges itself in mouse embryonic stem cell for functional copy resulting in the production of knockout mouse.

72-Transfer of Transgene in the Animal

The essence of this technology resides in successful gene transfer in the intended animal and its insertion in the genome. The insertion of gene in the genome can be random or targeted. There are several methods of producing transgenic animals:

(1) DNA microinjection,

(2) retrovirusmediated gene transfer,

(3) embryonic stem cell -mediated transfer, and

(4) liposomemediated DNA transfer into cells and embryos.

A few modifications are also there for creating transgenics:

- Electroporation of DNA into the sperm, ova, or embryos and
- Nuclear transfer with somatic or embryonic cells.

73-Pronuclear Microinjection

Microinjection was the first technique used to generate transgenic mice by successful gene transfer using SV40 DNA directly into blastocoel cavity of preimplantation embryos and then their implantation in the foster mother. Eventually, the gene of interest may randomly integrate in the embryonic genome of some embryonic cells and ultimately contributes to germline. The technique was then extended to other animals (livestock). The method has poor efficiency and usually results in a high percentage of mosaics in which not all cells of the animal contained the transgene. The time and cost of screening for germline transmission from the major mosaic animals prohibits the generation of more transgenic animals. The method can also lead to high variability in transgene expression between animals due to mosaicism, variable efficiency of transgene integration, and chromosomal position effects. The major advantage of this method is its applicability to a wide variety of species.

This method involves the direct injection of the gene construct into the cells to ensure guaranteed successful transfer of transgene to target cells. Presently the established technique involves the

injection of DNA into the pronuclei of the fertilized egg. The host cell is held in position with a suction pipette and DNA is transferred into the nucleus through a fine needle.

Thereafter the embryos with the transgene are cultured in vitro up to the morula stage and then transferred to pseudopregnant foster mothers. The insertion of DNA is, however, a complete random process; thus, there is high probability that either the introduced gene might be inserted into the host genome or, less commonly, it may remain extrachromosomal for one or more cell divisions. Thus, the resulting animal may be transgenic or chimeric for transgene insertion. Five to 40 % of mice developing from manipulated eggs contain the transgene. Once the transgene is transmitted through the germline, it tends to be stably inherited over many generations. The transgene insertion is accompanied by deletions and rearrangements occurring in genomic DNA. The limitation of microinjection is that it only allows for the random addition of exogenous DNA rather than targeting to specific sites. DNA targeting is necessary in generating gene knockouts, for instance, to model human diseases. The success of pronuclear microinjection is evident in the generation of transgenic pigs, goats, sheep, and cattle; but its limitations have hindered the progress of transgenics in livestock.

74-DNA Transfer Through Virus

Gene transfer mediated by means of a carrier or vector, generally, a virus or a plasmid, provides increased probability of expression. Virus has natural ability to stably introduce DNA to the host upon infection; therefore recombinant retroviruses and replication defective adenoviruses can be efficiently used as vectors. Retrovirus is efficiently capable of infecting dividing cells.

Initial attempts to produce transgenics by retroviral infection of early embryos invariably resulted in genetic mosaics caused by multiple insertion sites of the transgene. The short window of opportunity for the viral preintegration complexes to reach the embryonic chromatin during M-phase explains the delayed viral insertion, resulting in cell lineages with different insertion sites or no insertion at all. A significant advancement of this technique has been achieved recently by exposing metaphase II oocytes to transgenecontaining retrovirus. Arrested oocytes are particularly appropriate because they have undergone nuclear envelope breakdown and remain at metaphase II for a longer period compared with the M-phase of somatic cells (Fig.1). This maximizes the probability of preintegration complexes gaining access to the oocyte chromatin. The infected embryos are implanted in the uterus of foster mother to get germline transfer of transgene . Adenovirus represents an alternative to retrovirus as vectors to insert recombinant DNA into the mammalian germline.

However, the disadvantages of this technique are the possible interference of viral regulatory elements with the expression of genes and the susceptibility of virus to de novo methylation resulting in silencing of the genes. Though viruses are highly adapted to the process of gene transfer however, using virus as the vector may also produce health hazards as they may potentially arouse cancer/leukemia.

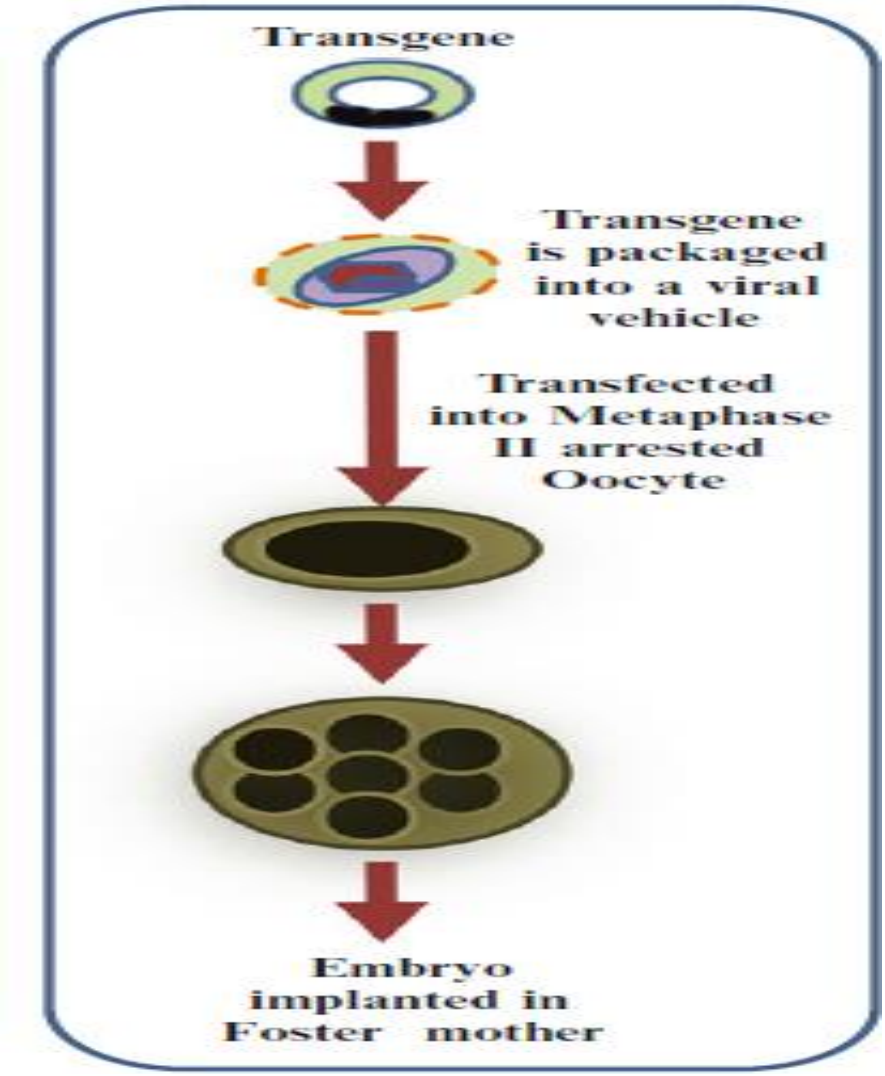


Fig-1 Virus-mediated transgenic production where the virus along with transgene is allowed to transfect metaphase II arrested oocyte. After transfection of oocyte, it is implanted in foster mother resulting in production oftransgenic animal

75-Embryonic Stem Cell Mediated Gene Transfer

Embryonic stem cells are obtained from inner cell mass of blastocyst. This method involves prior insertion of the desired DNA sequence by homologous recombination into an in vitro culture of embryonic stem (ES) cells, and the transformed cells can be selected using standard markers. Cells containing the desired DNA are incorporated into the blastocoel of host's embryo at the blastocyst stage, where they mix with inner cell mass (Fig.1). Unlike the other two methods,

which require live transgenic offspring to test for the presence of the desired transgene, this method allows testing for transgene at the cell stage. This is advantageous as there is no convenient way to select for eggs or embryos that have taken up foreign DNA.

The result is a true chimeric animal. ES cell-mediated gene transfer is the method of choice for gene inactivation, the so-called knockout method. This technique is of particular importance for the study of the genetic control of developmental processes. This technique works particularly well in mice. It has the advantage of allowing precise targeting of defined mutations in the gene via homologous recombination.

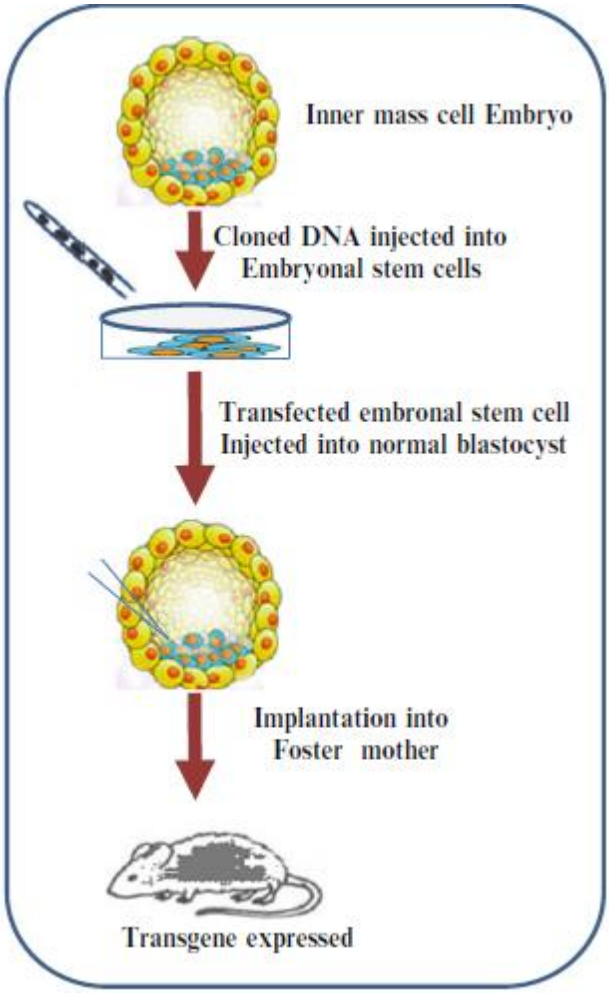


Fig- Embryonic stem cell-mediated gene transfer. The transgene is injected into the embryonal stem cells. These stem cells with transgene are injected into normal blastocyst, which is then implanted into the foster mother

76-Sperm-Mediated Transgenesis

The ability of sperm cells to carry exogenous DNA spontaneously into the oocyte during fertilization was first reported by Brackett and coworkers in 1971. However, the concept of sperm-mediated transgenesis rested for 18 years until scientists reported the use of spermatozoa as DNA carriers to produce transgenic mice. The scientific community was challenged by an unsuccessful attempt to replicate the experiment met this work with skepticism. However afterward transgene delivery by sperm cells was used to produce transgenic animals in a wide variety of species, including cattle, pigs, rabbits, frogs, and zebra fish. Sperm-mediated transgenesis is the most straightforward technique to produce transgenic animals. The spermatozoa are incubated with the DNA containing the gene of interest followed by in vivo or in vitro insemination.

DNA binds to the sperm's plasma membrane through specific DNA-binding proteins. After internalization, they are carried into the oocyte upon fertilization (Fig.1). The most appealing characteristics of using sperm as vectors to produce transgenic animals are its simplicity (no embryo manipulation is required) and the possibility of performing mass production of genetically modified animals through in vivo or in vitro insemination of many oocytes. On the other hand, this technique also has limitations.

Like pronuclear microinjection, no targeted modifications by homologous recombination could be achieved with this method due to the random integration of transgene. Recently scientists

reported successful generation of transgenic pigs carrying a human gene (human decay accelerating factor, hDAF). In this study, approximately 80 % of pigs had integrated the construct containing an hDAF minigene; from this 80 %, approximately 53 % expressed the foreign protein at different levels. There is no doubt that this success represents an important step toward production of humanized pig organs and tissues for human transplantation. If these findings are readily reproducible, sperm- mediated gene transfer may displace in short time the popular pronuclear microinjection method.

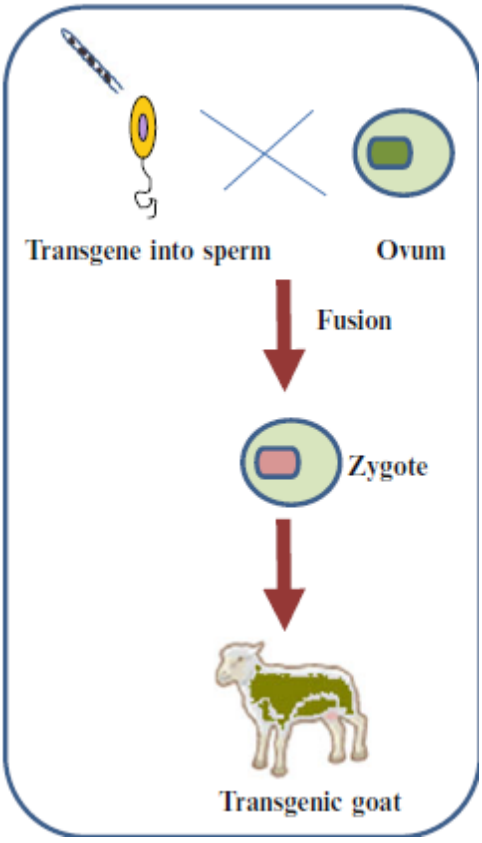


Fig-1 Sperm-mediated gene transfer where transgene is inserted into male pronucleus which upon fusion with ovum results in the production of transgenic goat

77-Somatic Cell Nuclear Transfer

Due to the absence of proven ES cells and the recent advances in nuclear transfer (NT), current emphasis for creating transgenic animals is on somatic cell nuclear transfer (SCNT). Nuclear transfer is a technique that is used to create a genetically identical copy, or a clone, of an animal. Nuclear transfer commonly involves the transfer or placement of a donor nucleus into the cytoplasm of an enucleated MII oocyte (Fig.1).

Donor cells can originate from a wide variety of cell types ranging from embryonic blastomeres all the way up to adult cells . Initial work on nuclear transfer was focused on using embryonic blastomeres as a donor source; the limited number of cells available in an early embryo hampered the process. Fetal or adult cells have been used successfully to clone all major livestock species, including sheep, cattle, goats, and swine. The ability to use cells where transgene can be introduced and which can be cultured increases the number of cells available to clone, thereby facilitating the ability to make transgenic animals.

Somatic cell nuclear transfer has facilitated the ability to make transgenic animals by circumventing most of the shortcomings of other transgenic techniques. First, the sex of the animal can be predetermined by choosing the donor material (i.e., male or female tissue). Second, the use of cell culture to propagate donor cells can lead to large numbers of transgenic cells that can be frozen and stored for long periods of time.

In conjunction with SCNT, these transgenic donor cells can eventually give rise to numerous cloned transgenic animals. Transgene structure and expression can be tested by molecular techniques, such as PCR , Southern blot analysis , fluorescence in situ hybridization, and Western blot analysis , before initiating nuclear transfer and transferring the embryo to a recipient cow with a lengthy gestation time of 9 months. The proper use of SCNT also ensures

that 100 % of animals produced are transgenic and that every cell of a cloned animal will have the transgene, thereby saving time and cost associated with recipient animals. The ability to use a clonal population of transgenic cells guarantees the same transgene insertion site for each clone, thus decreasing animal to animal variation in transgene expression levels. Further, transgene can be added to a specific genetic background. This technique can be used for the addition of DNA at random sites and targeted insertion of DNA by homologous recombination, which is vital in modulating specific gene expression as well as creating gene knockouts.

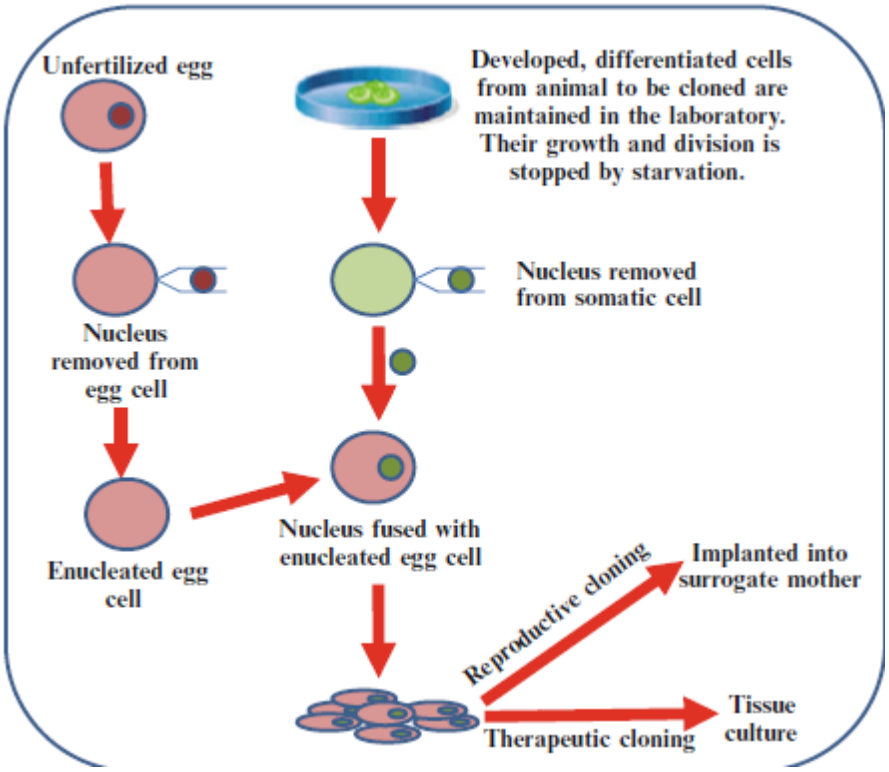


Fig-Somatic cell nuclear transfer where nucleus is removed from the egg cell and enucleated egg receives nucleus from donor somatic cell. The resultant egg with nucleus from somatic cell can be used for reproductive cloning or therapeutic cloning

78-Medical Applications of Transgenic Animals

In medicine the transgenic animals may serve as disease models, knockouts for studies related to gene function, quality trait improvement, and pharmaceutical production.

Transgenic Animals as Disease Model

Certain transgenic animals are produced for specific traits and serve as disease model. Animals are genetically manipulated to exhibit disease symptoms; thus they can be used for studies related to causes of the disease and effective treatment. For example, Oncomouse® developed by Harvard scientists is genetically modified and carries a gene that promotes the development of various human cancers.

Transgenic animals have enabled scientists to understand the role of genes in specific diseases. By either introducing or inactivating particular genes, researchers can discover the root causes of diseases associated with gene defects. Transgenic animals allow more effective treatments to be developed. Having found the genes implicated in a disease, scientists can then target these or design other therapies which act by influencing their expression.

Transgenic animals help test the safety of new medicines and vaccines. Because transgenic models can highlight specific characteristics such as certain mechanisms involved in the formation of tumors, they can demonstrate more clearly the possible side effects of new therapies. Their use in early toxicity trials may also serve to prevent the subsequent use of a larger number of animals in the development phase.

79-Xenotransplantation

The gap between the demand for and the availability of human organs for transplantation is growing, a tendency that is not likely to reverse in the near future. The number of patients on

waiting lists for organs has been growing every year, reflecting both an increased demand for organs for transplantation and a pronounced shortage of donors.

Transgenic animals are being developed by some companies to provide new organs for transplantation such as kidneys, livers, and hearts. Pigs have long been considered as an alternative source of organs for xenotransplantation (i.e., transplantation of organs/tissues between different species, e.g., from animals to humans) to satisfy this increasing demand. Pigs seem to meet most of the requirements for an ideal animal donor: they are anatomically and physiologically similar to humans with organs of appropriate size, they are prolific, and they can be maintained under specific pathogen-free conditions.

However, there are barriers to pig-to-human xenotransplantation; one of the main challenges is to avoid the hyperacute immunological rejection of the grafted tissue. When organs or tissues are transplanted between discordant species like pig to human, the host's immune system initiates a fast reaction known as hyperacute rejection (HAR).

80-Therapeutic proteins

One of the main discoveries in r-DNA technology was the production of biosynthetic human insulin, which was the first biomolecule made through r-DNA technology. Later on, this bioengineered insulin became the first biotechnology product approved by the U.S. Food and Drug Administration (FDA). It has been reported that insulin is basically an ideal candidate because it is a relatively simple protein and is therefore relatively easy to manipulate for genetic engineering.

The first step in making recombinant insulin is the introduction of a specific gene sequence (oligonucleotide) that codes for insulin production in humans into the bacteria *E. coli*; it has been reported that only 0.94% of bacteria can pick up the sequence. Nevertheless, because the life cycle of *E. coli* is only about 30 min, there is no problem in generating a million copies of *E. coli* in a short period of time to be used for inducing insulin production.

Among all applications of r-DNA technology, therapeutic proteins have been extensively used. Therapeutic proteins are those proteins that are either removed from human cells or engineered in the laboratory for pharmaceutical use. In addition to human insulin, other human proteins such as follicle-stimulating hormone, plasminogen, erythropoietin, and growth hormones have been created by using r-DNA technology (Figure 1).

There are many proteins essential to for the normal function of the human body and unfortunately some people fail to produce sufficient amounts of these proteins in their body, which may lead to the development of functional deformities and genetic defects. These essential proteins include various blood-clotting factors causing hemophilia, insulin, growth hormone, and other proteins. Patients who do not produce sufficient amounts of these proteins need to take these proteins externally either in the form of a medicine or an injection. These proteins can be synthesized outside of the human body by using recombinant technology. It has been reported that the majority of therapeutic proteins are recombinant human proteins manufactured using nonhuman mammalian cell lines that are engineered to express certain human genetic sequences to produce precise proteins of human use. Over the past decade, recombinant proteins are extensively used to replace deficiencies and to strengthen the immune system to fight cancer and infectious disease.

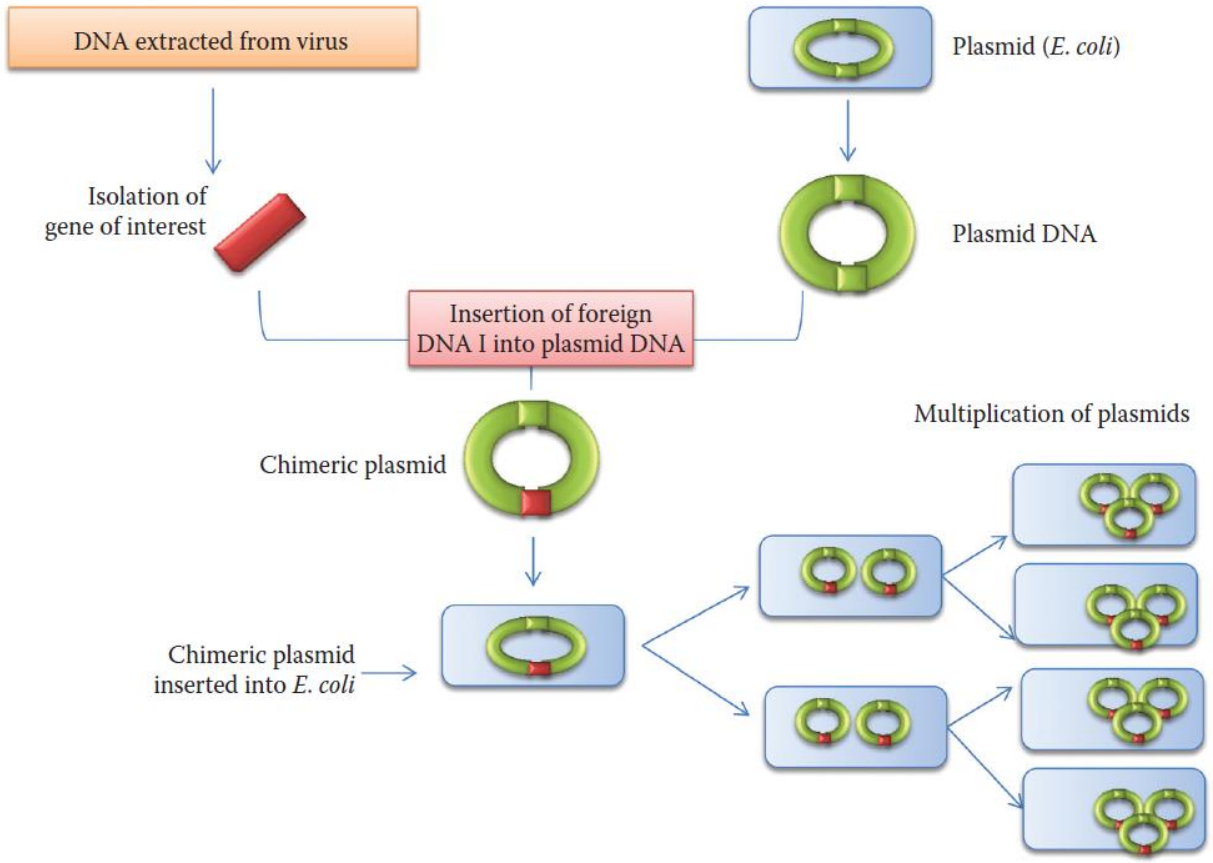


Fig-Making of chimeric proteins.

81-Recombinant human insulin

Insulin is a peptide hormone consisting of 51 amino acids. It is secreted by β cells of islets of Langerhans of pancreas. The hormone is responsible for maintaining normal blood glucose level in blood. Insulin is stored in the form of proinsulin which contains two polypeptide chains, A and B, and is connected with a third peptide C chain), which before secretion is cleaved with production of insulin and C-chain. The cleavage results in the removal of C-chain, and the A (21 amino acids) and B chain (30 amino acids) are linked by disulfide linkage to form mature insulin.

In the beginning, efforts were made to isolate mRNA for pre- and proinsulin from rat islets of Langerhans of pancreas and to synthesize cDNA. Thereafter, it was inserted into a plasmid. The recombinant plasmids were transferred into the *E. coli* cells, which secreted proinsulin. Scientists have chemically synthesized DNA sequences for two chains, A and B, of insulin and separately inserted into two pBR322 plasmids by the side of β-galactosidase gene.

The recombinant plasmids were separately transferred into *E. coli* cells which secreted fused β galactosidase-A chain and β-galactosidase-B chain separately. These chains were isolated in pure form by detaching from β-galactosidase with yields of about 10 mg/24 g of healthy and transformed cells. Production of recombinant insulin is shown in (Fig.1).

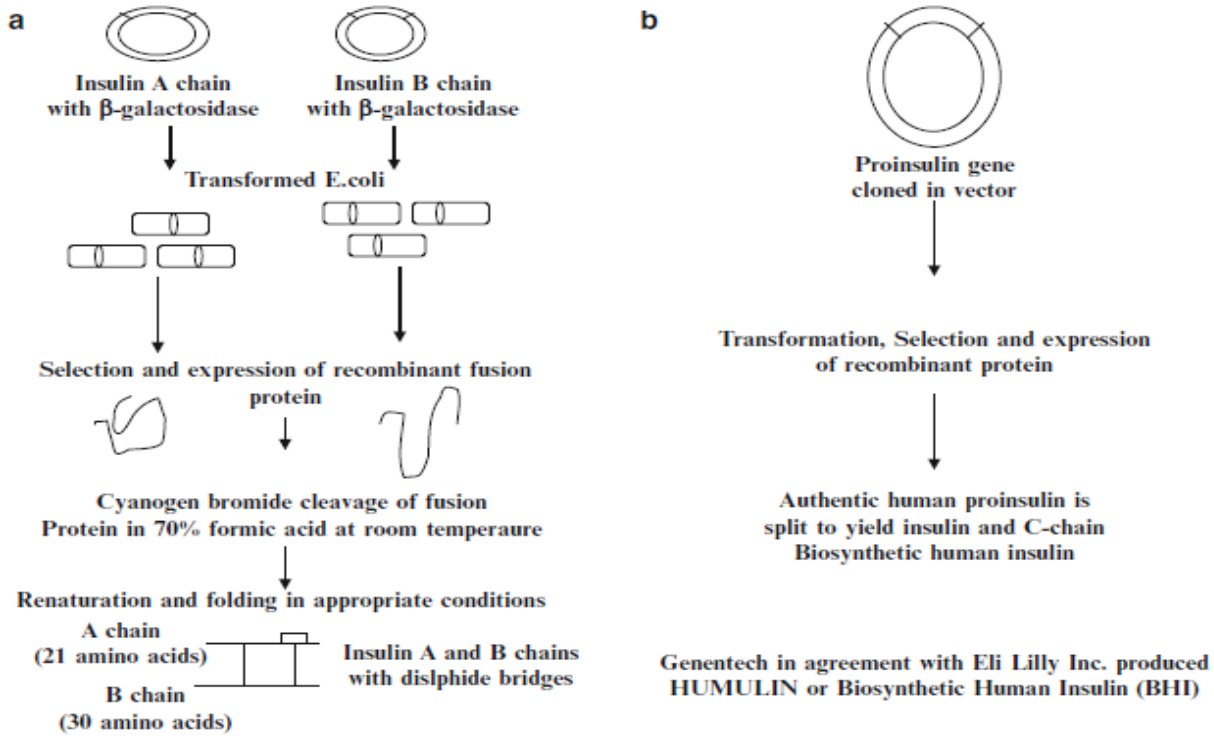


Fig-The fi gure shows the production of insulin. (**a**) The A and B chains of insulin are cloned separately in the vector. The transformation is done and after selection, the recombinant protein is obtained for both A- and B-chain genes. Cleavage by using cyanogen bromide removes the bacterial protein. The protein is renatured and provided suitable conditions for folding. (**b**) This figure shows the production of insulin from proinsulin gene which yields insulin and C-peptide. This is preferred method for production of biosynthetic human insulin.

82-Recombinant human growth hormone

Growth hormone is produced by the anterior lobe of the pituitary gland and is released in multiple pulses. The hGH is encoded by GHN gene cluster (an array of fi ve closely related genes), which is localized on chromosome 17. It belongs to diverse gene family that has evolved by gene duplication events and has lots of structural similarity and some common functions. Of the various forms, the predominant form of hGH is 22 kDa protein of 191 amino acids with two disulfide bonds.

GH does not control the functions directly, but acts on certain hormones or somatomedins for its activity, for example, insulin-like growth factor 1 (IGF-1). It has a wide spectrum of roles to play as promotion of long bone growth, promotion of normal sex organ development through puberty, regulation of metabolism, stimulation of tissue growth and repair, anabolic/anti-catabolic effect via improved nitrogen retention, modulation of bone mineral density and metabolism throughout life, proliferation of some cell types of the immune system, appetite stimulation, and breakdown of fat (lipolysis). In clinical conditions, the therapy of growth hormone is given in the treatment of dwarfism, bone fractures, skin burns, bleeding ulcers, and AIDS.

Recombinant human growth hormone (rhGH) is 22 kDa consisting of long chain of amino acids. It is used in deficiency disorders of growth hormone. In children, it is used for growth abnormality as short stature and is used in chronic renal insufficiency. The therapy of growth hormone is also approved for adult growth hormone deficiency. GH is one of the most widely used hormones with the estimated market of more than 1.7 billion USD. Long experience in its

administration has proven the therapy as safe and effective in various conditions of growth abnormality.

Earlier pituitary-derived hGH was used but later on it was prohibited when found associated with Creutzfeldt–Jakob disease. Because of recombinant DNA technology, safe and abundant recombinant hGH was produced in various heterologous systems. As the non-glycosylated human growth hormone was biologically active, thus, the preferred system for its production is *E. coli* , which allows its rapid and economical production in large amounts.

Recombinant hGH (rhGH) is now used to treat:

- GH-deficient (GHD) short-stature children.
- Acceleration of wound healing.
- Increase in insulin-like growth factor (IGF)-1 levels.
- rhGH increases IGF-1, osteocalcin, type I procollagen pro-peptide (PICP), and bone density, when administered to children with GHD

For optimal productivity, strong inducible promoters are preferred as IPL, IPR, trc, and T7 in *E. coli*. They are advantageous as they drive overproduction of recombinant proteins. Apart from *E. coli*, human somatotropin (hST) expression was tried in a biologically active, disulfide- bonded form in tobacco chloroplasts. The hormone is used for the treatment of hypopituitary dwarfism in children; additional indications are in treatment of Turner syndrome, chronic renal failure, HIV wasting syndrome, and possibly treatment of the elderly. Growth hormone deficiency in human occurs both in children and adults.

83-Recombinant blood clotting factor VIII

Factor VIII is one of the important factors of all blood clotting factors. Deficiency of factor VIII causes bleeding disorder called hemophilia A. The hemophilia may be mild to severe depending upon factor VIII concentration in the body. In moderate and severe factor VIII deficiency, there can be spontaneous bleeding episodes in the joints. Hemophilia A affects 1 in 5,000–10,000 males. Replacement therapy is the treatment option for hemophiliacs either with human plasma-derived factor VIII (pdFVIII) or recombinant FVIII (rFVIII).

Transfection of HEK 293 cell cultures in serum-free suspension is being tried for optimal yield. Recombinant factor VIII (rFVIII) is produced by culturing mammalian cells as baby hamster kidney (BHK) or Chinese hamster ovary cells (CHO), using large-scale bioreactors . Standardizations are done to maximize yields.

84-Recombinant tissue plasminogen activator

Tissue plasminogen activator (tPA) is another kind of serine protease protein that is involved in the breakdown of blood clots in humans (Figure 1). In the human body, tissue plasminogen is found on endothelial cells, the cells that line the blood vessels. As an enzyme, tPA catalyzes the conversion of plasminogen to plasmin, the major enzyme responsible for clot breakdown.

It has been reported that tPA is used as clinical medicine to treat only embolic or thrombotic stroke. Patients who have a problem related to blood clotting need a medicine to clot the blood immediately to avoid further releases of the blood from the body and tPA is considered to be an important medicine to promote blood clot. The tPA can be manufactured by using r-DNA

technology and the production of tPA created this way is referred to as recombinant tissue plasminogen activator (rtPA).

The basic principle behind blood clotting is that tPA catalyzes the conversion of plasminogen into plasmin, which is done by cleaving the single-chain plasminogen into two chains. These two chains are connected by a disulfide bond and the resulting molecule is called plasmin. In the patients, an increased enzymatic activity can cause hyper fibrinolysis, which demonstrates as an excessive bleeding.

tPA is used in various diseases such as pulmonary embolism, myocardial infarction, stroke, and thrombolysis. The first step in any kind of blood-clot-related disease is to stop the outflow of the blood, so the first thing any physician will do is to immediately inject tPA within the first 3 h of the event or within 6 h to be able to clot the blood. As per the guidelines used in Canada, hospitals for ischemic strokes suggest that tPA must be given within 4.5 h of the onset of symptoms, and because of this, only about 3% of patients were able to qualify for this treatment, as most patients do not seek medical assistance quickly enough. With regard to the situation in the United States, the window of the administration of tPA is 3 h from the onset of symptoms

There are various commercially available rtPAs in the market, which include Alteplase, Reteplase, and Tenecteplase. It has been reported that Alteplase is an FDA-approved drug for the treatment of myocardial infarction, acute ischemic stroke, and acute massive pulmonary embolism. Reteplase is also an FDA-approved drug for the treatment of acute myocardial infarction and it has more convenient administration and faster thrombolysis than Alteplase.

Consequently, Tenecteplase is also specified in acute myocardial infarction, and it showed fewer bleeding complications. There are other types of tPAs are under trial phases, including Desmoteplase. One of the major concerns with regard to tPA treatment is that tPA quickly dissolves blood clots, which might lead to a risk of hemorrhage in some patients. Considering the fact that the use of tPA may cause hemorrhage, there are alternative medicines available in the market that do not cause any hemorrhage. One example is streptokinase, which is a cheaper alternative that can be used as a thrombolytic in acute treatment.

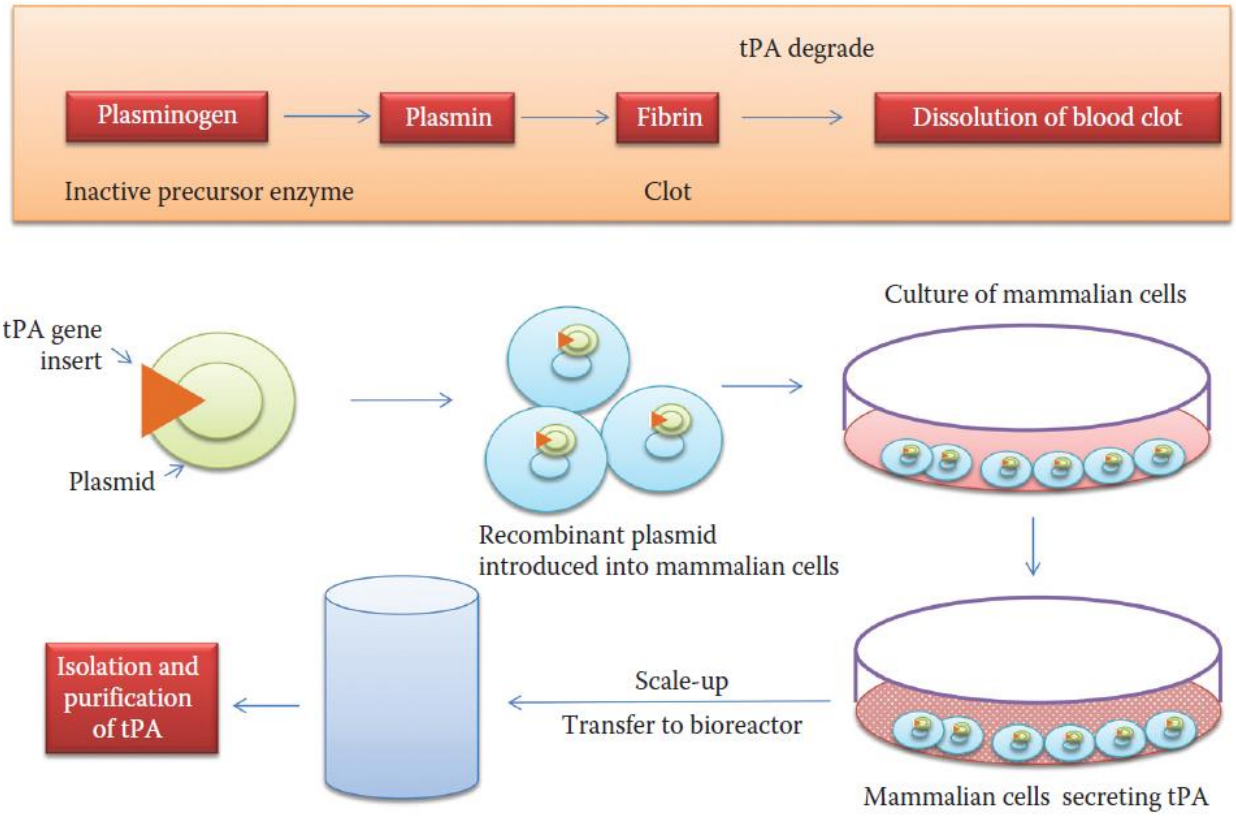


Fig-1 Making of recombinant tissue plasminogen activator.

85-Recombinant erythropoietin

Like insulin, erythropoietin (EPO) is a glycoprotein hormone that controls red blood cell production in the body (Figure 1). EPO, also called hemopoietin, is known to be located as precursors in the bone marrow. It is produced by the peritubular capillary endothelial cells in the kidney and liver. Besides its role in red blood cell production, EPO also plays an important role in neuronal injury and in the wound healing process. It has been reported that EPO has its primary effect on red blood cells by protecting the blood cells from going to the apoptosis pathway. It also conjoins with various factors involved in the growth of precursor red cells. EPO has a range of actions in hypertension, stimulating angiogenesis, and inducing proliferation of smooth muscle fibers. It has also been shown that EPO can upsurge iron absorption by quashing the hormone hepcidin.

Applications of erythropoietin

There are various applications of EPO. EPO can be used as a therapeutic agent to treat various disease conditions. It can be used in the treatment of anemia resulting from chronic kidney disease and myelodysplasia, and in the treatment of cancer. There are various types of EPO produced for patients who are suffering from anemia. These are epoetin, darbepoetin, epoetin delta, PD poetin, and methoxy polyethylene glycol-epoetin beta. The list of erythropoietin produced by using r-DNA technology is given in Table -1.

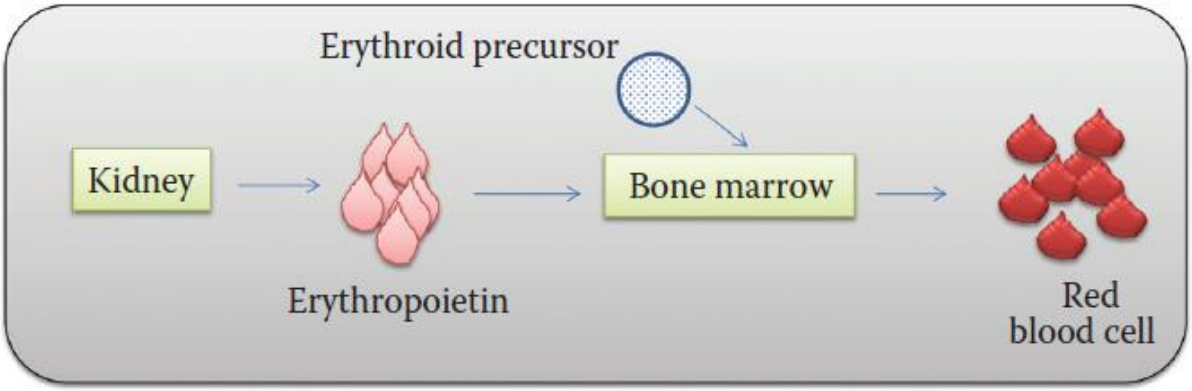


Fig-Normal synthesis of red blood cells

Table-List of Erythropoietin Produced by r-DNA Technology

Trade name of erythropoietin	Name of manufacturer
Epogen	Amgen
Espogen	LG Life Sciences
Epotin	Gulf Pharmaceutical Industry (Julphar)
Betapoietin	CinnaGen and Zahravi
Erykine	Intas Biopharmaceutical Pvt. Ltd.
Shanpoietin	Shantha Biotechnics Ltd.
Zyrop	Cadila Healthcare Ltd.

86-Significance of vaccines

A vaccine is a biological preparation that improves human immunity to a particular disease. Moreover, a vaccine basically contains an agent that resembles a disease-causing microorganism, and is often made from weakened or killed forms of the microorganism or its toxins. Upon injecting these vaccines in humans, the vaccine molecules stimulate the body's immune system to recognize the agent as foreign, destroy it, and memorize it, so that the immune system can more effortlessly familiarize with it and destroy any of these microorganisms that it later encounters. The story behind vaccine development started with the research work done by Edward Jenner in the late 1700s. Jenner took pus from the hand of a milkmaid with cowpox, inoculated an 8-year-old boy with it, and 6 weeks later variolated the boy's arm with smallpox,

afterwards observing that the boy did not catch smallpox. Later on, more experimentation was performed to demonstrate the efficacy of the procedure in infants. Interestingly, Louis Pasteur generalized Edward Jenner's idea by developing what he called a rabies vaccine, now termed an antitoxin, which caused an extensive development of vaccines around the globe and obligatory vaccination laws were passed. Furthermore, researcher Maurice Hilleman was the most successful developer of vaccines in the twentieth century. Vaccines became more common among all populations of the world; however, vaccines remain elusive for many important diseases, including malaria and HIV.

Type of vaccines

Vaccines are produced by either dead or inactivated organisms and there are four main ways to develop vaccines: (i) live attenuated vaccines contain bacteria or viruses that have been altered so they cannot cause disease, (ii) killed vaccines contain killed bacteria or inactivated viruses, (iii) toxoid vaccines contain toxins (or poisons) produced by the germ that have been made harmless, and (iv) component vaccines contain parts of the whole bacteria or viruses.

87-Live attenuated vaccines

The live attenuated or weakened vaccines can be created from the naturally occurring microorganisms; however, microorganisms used in this vaccine development can still infect people, but they rarely cause severe disease. Viruses are weakened by growing them over and over again in cell culture conditions. Furthermore, the procedure of growing a virus under cell culture conditions is known to decrease their disease-causing capabilities. There are various examples of live attenuated vaccines that include measles vaccine (as found in the MMR

vaccine), rubella (German measles) vaccine (MMR vaccine), oral polio vaccine (OPV), mumps vaccine (MMR vaccine), and varicella (chickenpox) vaccine.

The advantages of attenuated vaccines are:

• Single dose is effective, and boosters are not essentially required because immune system is exposed to the live organism for prolonged period.

• Efficiently activates adaptive immune responses.

Disadvantages:

• Storage conditions are to be strictly maintained.

• Reversion is one of an important risk.

• Vaccines can cause serious complications in immunocompromised patients

88-Inactivated vaccines

Inactivated vaccines cannot cause an infection, but they still can stimulate a protective immune response. Due to problem of reversion of pathogenic agent in attenuated vaccines, the live virus/bacteria were inactivated by heat or chemical treatment. Viruses can be inactivated with chemicals such as formaldehyde. These inactivated vaccines have microorganisms incapable of replication but is able to activate immune system for memory response. Controlled inactivation is done for production of vaccine where microorganism is properly inactivated (that is killed), but at the same time, care is taken to maintain the surface structure of epitope (for specificity and memory). Heat inactivation is not preferred because it causes denaturation and thus loss of native structure of the epitopes. Chemical inactivation using formaldehyde or various alkylating agents has been successfully used. In contrast to attenuated vaccine where one dose is sufficient for long term immunity, inactivated killed vaccines require boosters with appropriate adjuvants for activating immune system of the host. The humoral immune activation is the predominant

immune system branch, which responds for inactivated/ killed vaccines. One example of an inactivated vaccine is the inactivated polio vaccine (IPV), which is the shot form of the polio vaccine, inactivated influenza vaccine.

The complications of inactivated vaccines were observed when Salk vaccine for polio was prepared and formaldehyde failed to kill the entire virus in two lots resulting in outbreak of paralytic polio in a high percentage of recipients.

89-Toxoid vaccines

There are some kinds of vaccines that can be made by treating them with toxins (formalin) produced by microorganisms, to destroy their ability to cause illness. Additionally, toxoids do not cause disease; in fact, they stimulate the body to produce protective immunity. There are several examples of toxoid-based vaccines such as diphtheria toxoid vaccine, which may be given alone or as one of the components in association with DTP, and furthermore, tetanus toxoid vaccine can be given alone or as part of the DTP.

Component vaccines

Some vaccines are made by using only parts of the viruses or bacteria. These vaccines cannot cause disease; nevertheless, these vaccines can stimulate the body to create an immune response that protects against viral infection with the whole germ. So far, four vaccines have been produced by using this method that includes hemophilic influenza type b (Hib) vaccine, hepatitis B (Hep B) vaccine, hepatitis A (Hep A) vaccine, and pneumococcal conjugate vaccine.

90-DNA Vaccines

DNA vaccines came into being in 1990. Since that time they have been hailed as a novel approach to DNA technology. Entire scientific meetings are devoted to DNA vaccines, an Internet site has been created to disseminate germane information, and according to the respected journal Science "... biotech companies are tripping over each other to get into the business..." of producing DNA vaccines

DNA vaccines consist of plasmids engineered to contain a protein-encoding gene and all the necessary requisites. Unlike replicating viruses or live bacteria, they are not infectious or replicative, nor do they encode any proteins other than those specified by the plasmid genes. Thus, one of the outstanding features of the DNA vaccine is its safety.

But a DNA vaccine is not simple: A single plasmid might contain a promoter site, a convenient cloning site for the gene of interest, a poly-A tail used as a termination sequence, an origin of replication, and a selectable marker sequence such as an ampicillin-resistant gene (in addition, of course, to the protein-encoding gene or genes).

One highlight of the DNA vaccine is that no special formulation is necessary. Animals can be immunized against disease merely by injecting plasmids suspended in saline (salt) solution. And delivery of the plasmids to target cells is relatively easy: as early as 1994, researchers identified five effective ways for delivering influenza virus plasmids to the animal body: by nasal spray; by injection into the muscle, vein, and under the skin; and by a so-called gene gun. The latter is a propulsion device that shoots DNA-coated gold beads into the skin. All techniques resulted in protective immune responses.

Furthermore, deploying plasmids as DNA vaccines has the added advantage of stimulating both antibody-mediated immunity (related to B-lymphocytes) and cell mediated immunity (related to T-lymphocytes). This is because DNA vaccines encode proteins that are released from the cell and taken up by phagocytes for delivery to the B-lymphocytes for antibody production. At the same time, some of the proteins fix themselves to the target cell surface and stimulate a response by T lymphocytes.

Later when the pathogen appears, the antibodies and T-lymphocytes will still be available to neutralize the pathogen's proteins and destroy the pathogen itself (or the cell harboring the pathogen). DNA vaccines also appear to elicit a higher immune response than scientists anticipated, despite the tiny amount of antigenic protein encoded. Researchers guess that this added benefit is due to an immune response to the other genes contained in the plasmid. The products encoded by these genes, like the main gene products, are interpreted as foreign and "...increase the wallop of a DNA vaccine," as one writer has said.

Even the production of DNA vaccines is advantageous. The technological methods for producing the plasmids are now in place, and biochemists can adapt their methods for various DNA

vaccines. By using similar fermentation, purification, and validation techniques to produce a spectrum of DNA vaccines, manufacturers can reduce time and cost. Moreover, DNA vaccines are more stable at low and high temperatures compared with conventional vaccines. They can be shipped easily and inexpensively and reach a greater population of vaccinees.

At this writing, DNA vaccines have been used experimentally to protect against influenza, Salmonella typhi, HIV infection, herpes simplex, hepatitis B, and several cancers including melanoma, colon cancer, renal cell carcinoma, and T-cell lymphoma.

Public health specialists envision that using DNA technology, vaccines can be developed quickly against emerging diseases (e.g., those caused by Ebola virus and drug-resistant bacteria), and they foresee that a number of different diseases can be controlled with a single vaccination by simply adding more genes to the plasmid.

There are drawbacks to consider, however. The DNA vaccines do little to address pathogenic mechanisms involving carbohydrates (as in streptococcal pneumonia, for instance). Delivery systems must be worked out, and ages determined for optimum immune response. The vaccines must also be evaluated for therapeutic uses in already infected patients and in those with chronic conditions.

Over the years, the research with DNA vaccines has been modified and expanded for various purposes. One approach has been to induce the body's B-lymphocytes to produce both antigens and antibodies. In 1997, researchers Stephen Johnston and Maurizio Zanett| at the University of California described how they induced a mouse's B-lymphocytes to perform double duty: The researchers injected the B-lymphocytes with plasmids containing genes from malarial parasites and genes that encode antibodies against malarial parasites. Soon the spleen cells were producing antibody molecules bound to antigens from the malarial parasite. The antibodies reacted with the malarial parasite causing disease while the antigen portions stimulated other B-lymphocytes to produce still more antibodies. The "antigenic antibody" is an extremely novel idea in immunology.

Researchers at the University of Texas have been equally imaginative as they attempt to identify which plasmids are the key to defense against a pathogenic organism (in effect they hunt for the plasmid giving the strongest immune response).

The scientists take a pathogenic organism's DNA and break it to fragments. Then they prepare perhaps 1000 different plasmids. Various plasmid groups are injected to animals as DNA vaccines, and the animal is given time to develop an immune response. Then the pathogen is injected. If the animal remains unaffected by one of the plasmid groups, the key element for protection is in that group. The group is subdivided into smaller groups and the test is repeated. Again, the uninfected animal points to the smaller plasmid group. Retest after retest follows, and the scientists eventually reach a single or few plasmids that are conferring protection. Those are the ones presumably useful for a DNA vaccine.

The fresh approaches to vaccine research bode well for future generations and represent an elegant application of the DNA technology. Scientists have had difficulty restraining their enthusiasm for the new vaccines. But that is not unusual because they have had difficulty restraining their enthusiasm for most prospects of DNA technology.

The advantages are:

Protein is expressed in its native conformation.

Long-term exposure of immune system to the antigen.

Efficient activation of cell-mediated and humoral immune systems leading to memory response.

There is no requirement of specific storage condition like refrigeration of plasmid DNA, and it reduces the cost and is easy to handle.

The genes expressing cytokines can be ligated to the vector, and the cytokine produced at the site may activate the desirable branch of immune system.

Vaccine is safe, without any infectious agent.

Easier to produce in large quantity.

The disadvantage of DNA vaccine is that they can be used for only protein antigens, and for the non-protein antigens (polysaccharide) they cannot be used.